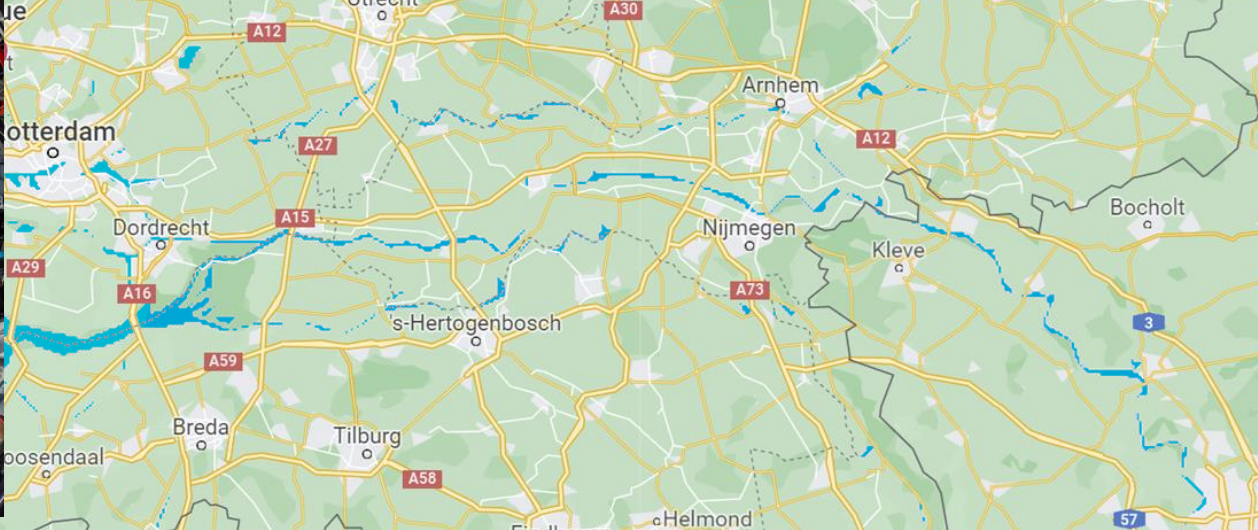


Fundamentals of Software Systems

Intro to Data Engineering for AI Systems

Dr. Christoph Lofi











The Modern World produces a humongous amount of Data



AI is one of the Current Tech Trends



“Artificial intelligence (AI) will be the most powerful technology in generations for benefiting humanity.”

“Artificial Intelligence (AI) has the potential to solve some of **society's most important challenges**, for example the mounting pressure on **healthcare**, while at the same time enriching and improving many other aspects of our society, such as **transport**. In short, the impact of AI is enormous, and it may even become the foundation of our future.”

– from NWO LTP ROBUST

Hype Cycle for Emerging Technologies, 2020



gartner.com/SmarterWithGartner

Source: Gartner
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Gartner

But...

- ...how successful are we really with using AI?
 - ...and especially with the ever-popular Deep Learning? (DL)
- AI is not magic - it is an incredibly powerful tool for extracting regularities from data according a given objective.
 - Claim #1: An AI system will be just as smart as the data it gets.
 - Claim #2: An AI system will be just as smart as the objective it optimizes.

Challenges

- Many open challenges...
 - Where does data come from?
 - Data Quality
 - The right training data?
 - Training Data Balanced?
 - Model Evaluation
 - “Weird Features”
 - Common Sense Reasoning
 - Explanations



“To err is human’ but a human error is nothing to what a computer can do if it tries.”

from Agathe Christie “Hallowe’en Party”, 1969

“People worry that computers will get too smart and take over the world, but the real problem is that they're too stupid and they've already taken over the world.”

from Pedro Domingos “The Master Algorithm”, 2015

The “hidden depth” of ML

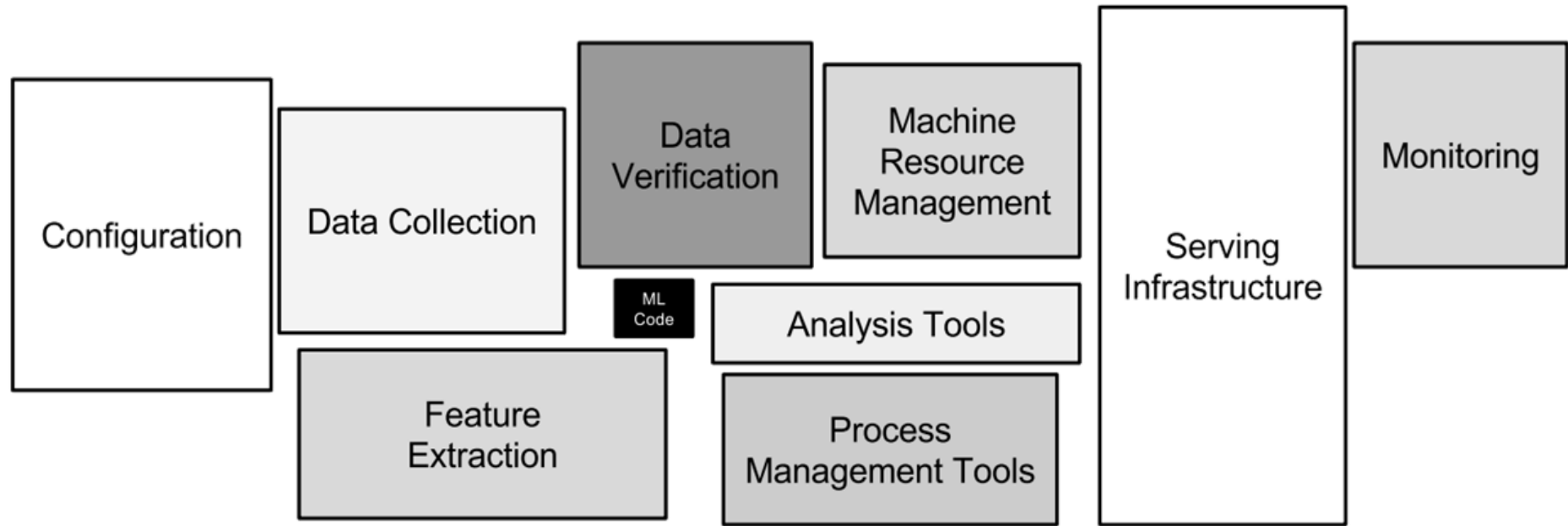
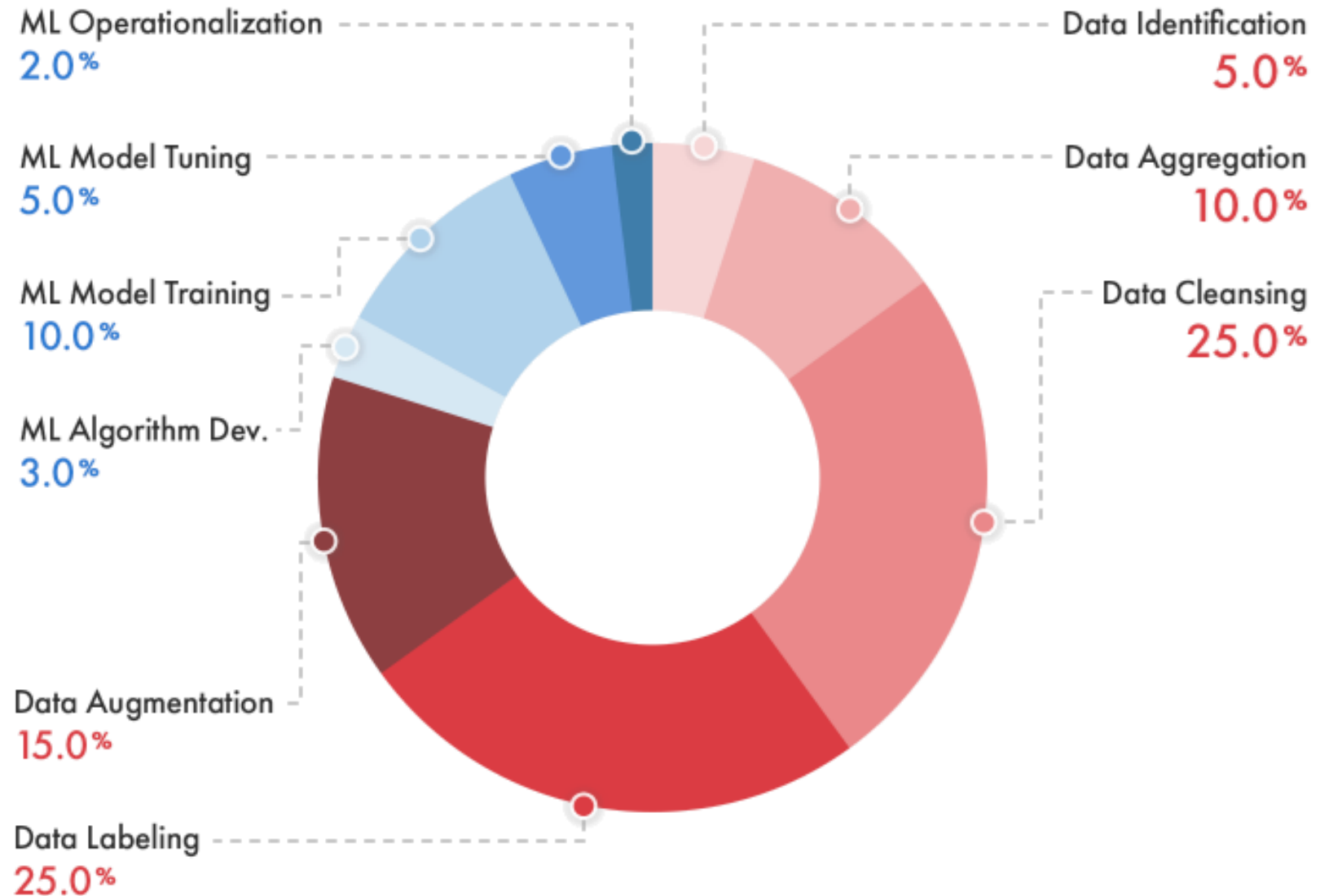


Figure 1: Only a small fraction of real-world ML systems is composed of the ML code, as shown by the small black box in the middle. The required surrounding infrastructure is vast and complex.

Percentage of Time Allocated to Machine Learning Project Tasks



A conceptual image featuring a large iceberg floating in a blue ocean under a sky with white clouds. The iceberg's tip is above the water line, while its much larger, jagged base is submerged. The text 'AI' is positioned above the water line, and 'Data Management' is positioned below it, both in white. The entire image is framed by a blue background with abstract geometric shapes.

AI

Data Management

AI and Big Data

Putting Big Data to use?



“All you’re seeing now — all these feats of AI like self-driving cars, interpreting medical images, beating the world champion at Go and so on — these are very narrow intelligences, and they’re really trained for a particular purpose. They’re situations where we can collect a lot of data.”

Yann LeCun, Facebook Head of AI, 2017

40 ZETTABYTES

[43 TRILLION GIGABYTES]
of data will be created by 2020, an increase of 300 times from 2005

6 BILLION PEOPLE
have cell phones



WORLD POPULATION: 7 BILLION

Volume SCALE OF DATA

2005

2020

It's estimated that
2.5 QUINTILLION BYTES
[2.3 TRILLION GIGABYTES]
of data are created each day

Most companies in the U.S. have at least
100 TERABYTES
[100,000 GIGABYTES]
of data stored



The FOUR V's of Big Data

From traffic patterns and music downloads to web history and medical records, data is recorded, stored, and analyzed to enable the technology and services that the world relies on every day. But what exactly is big data, and how can these massive amounts of data be used?

As a leader in the sector, IBM data scientists break big data into four dimensions: **Volume, Velocity, Variety and Veracity**

Depending on the industry and organization, big data encompasses information from multiple internal and external sources such as transactions, social media, enterprise content, sensors and mobile devices. Companies can leverage data to adapt their products and services to better meet customer needs, optimize operations and infrastructure, and find new sources of revenue.

By 2015
4.4 MILLION IT JOBS
will be created globally to support big data, with 1.9 million in the United States



As of 2011, the global size of data in healthcare was estimated to be

150 EXABYTES
[161 BILLION GIGABYTES]



**30 BILLION
PIECES OF CONTENT**
are shared on Facebook
every month



Variety DIFFERENT FORMS OF DATA

By 2014, it's anticipated there will be
**420 MILLION
WEARABLE, WIRELESS
HEALTH MONITORS**

**4 BILLION+
HOURS OF VIDEO**
are watched on
YouTube each month



400 MILLION TWEETS
are sent per day by about 200
million monthly active users



The New York Stock Exchange captures

**1 TB OF TRADE
INFORMATION**
during each trading session



By 2016, it is projected there will be

**18.9 BILLION
NETWORK
CONNECTIONS**

— almost 2.5 connections
per person on earth

Velocity ANALYSIS OF STREAMING DATA



Modern cars have close to
100 SENSORS
that monitor items such as
fuel level and tire pressure



**1 IN 3 BUSINESS
LEADERS**

don't trust the information
they use to make decisions



Poor data quality costs the US
economy around
\$3.1 TRILLION A YEAR



**27% OF
RESPONDENTS**

in one survey were unsure of
how much of their data was
inaccurate

Veracity UNCERTAINTY OF DATA

Let's time travel a decade..

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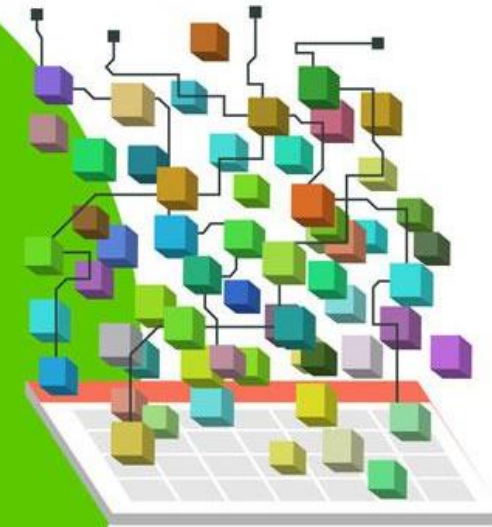


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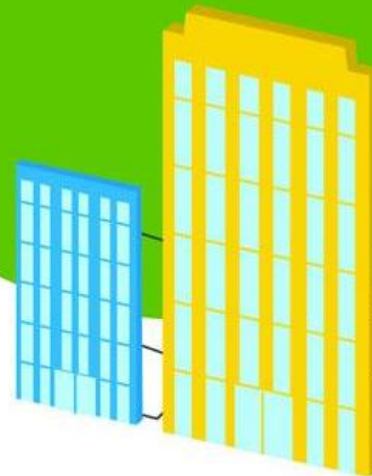


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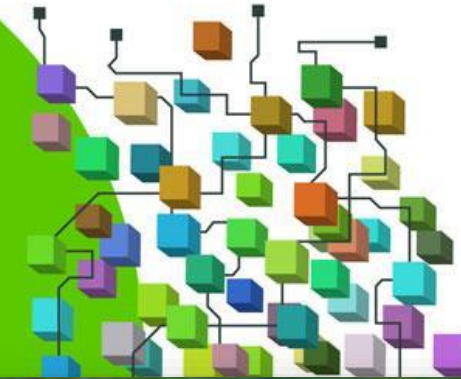


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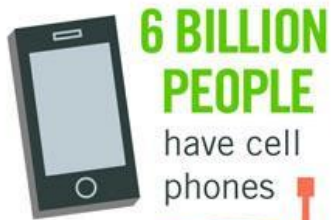
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Volume
SCALE OF DATA



WORLD POPULATION: 7 BILLION

WHARRGARBL



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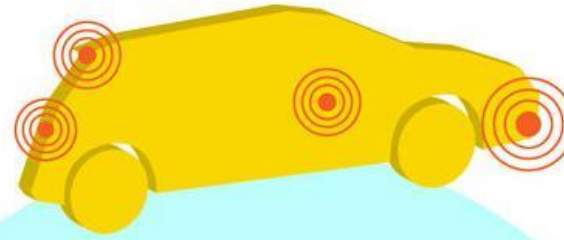
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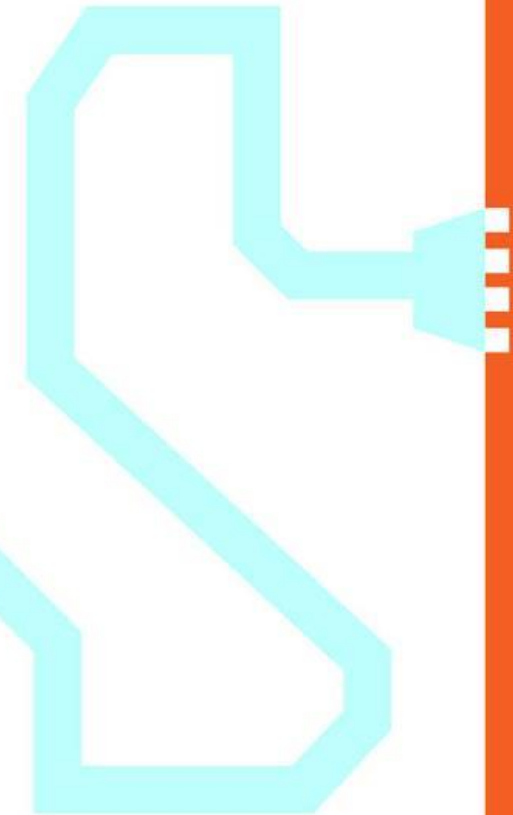


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Velocity

ANALYSIS OF STREAMING DATA



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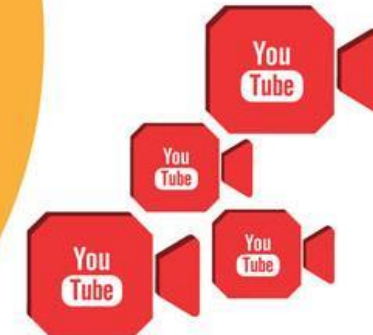
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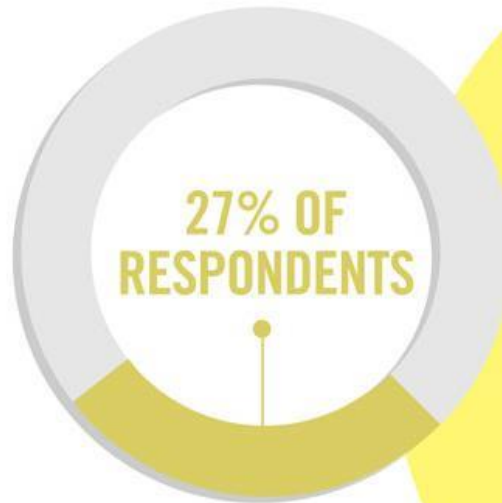
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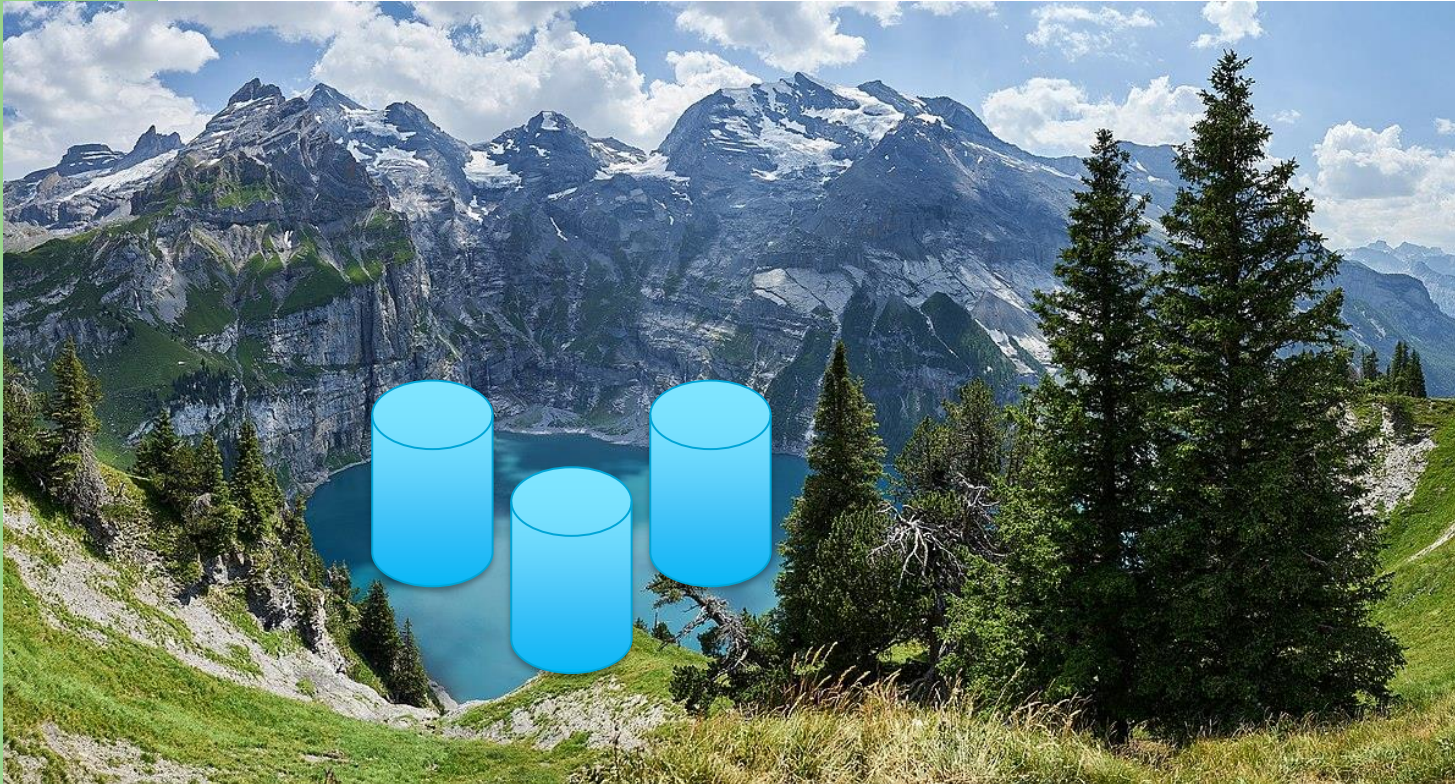
Veracity

UNCERTAINTY OF DATA

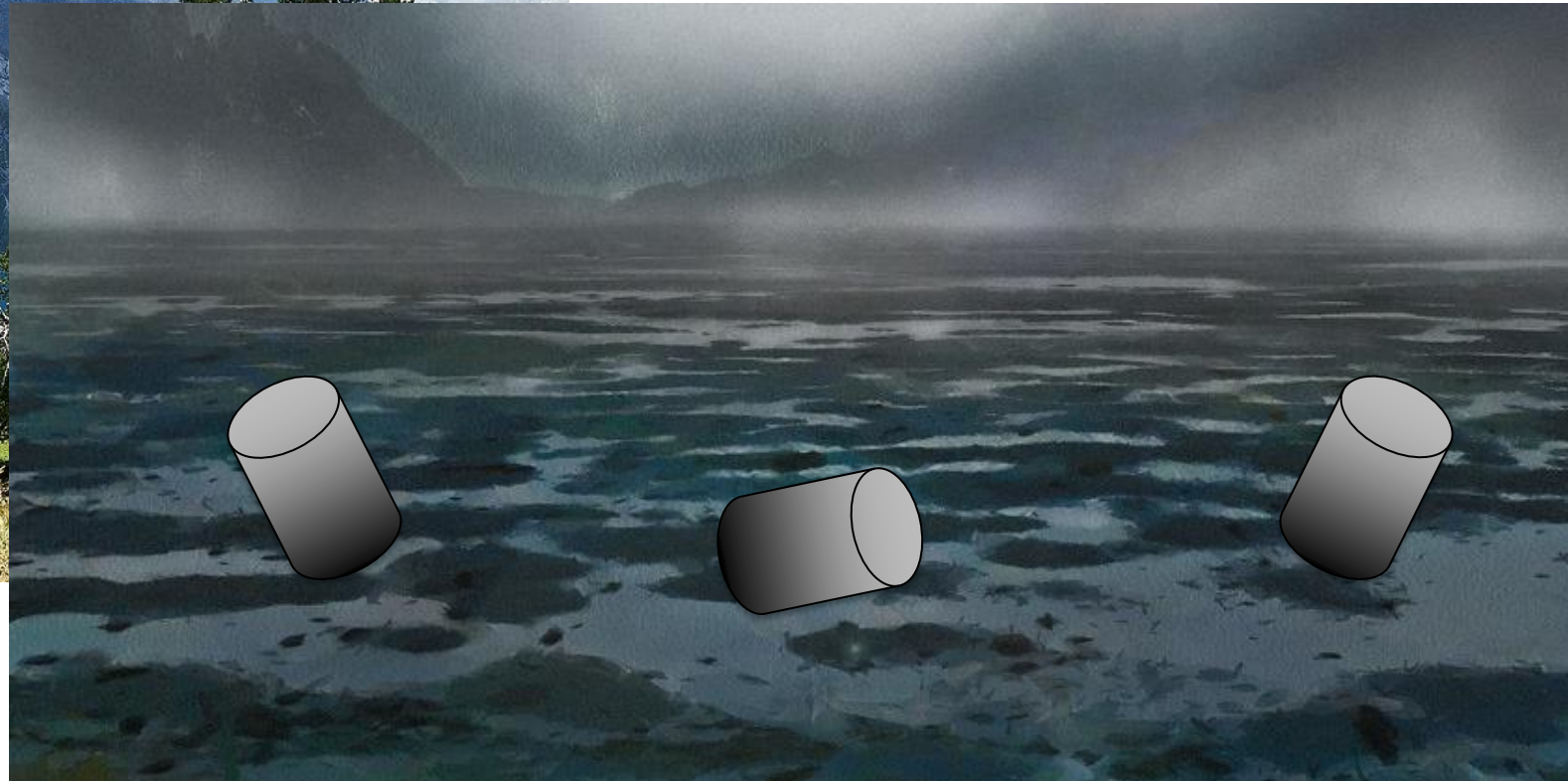
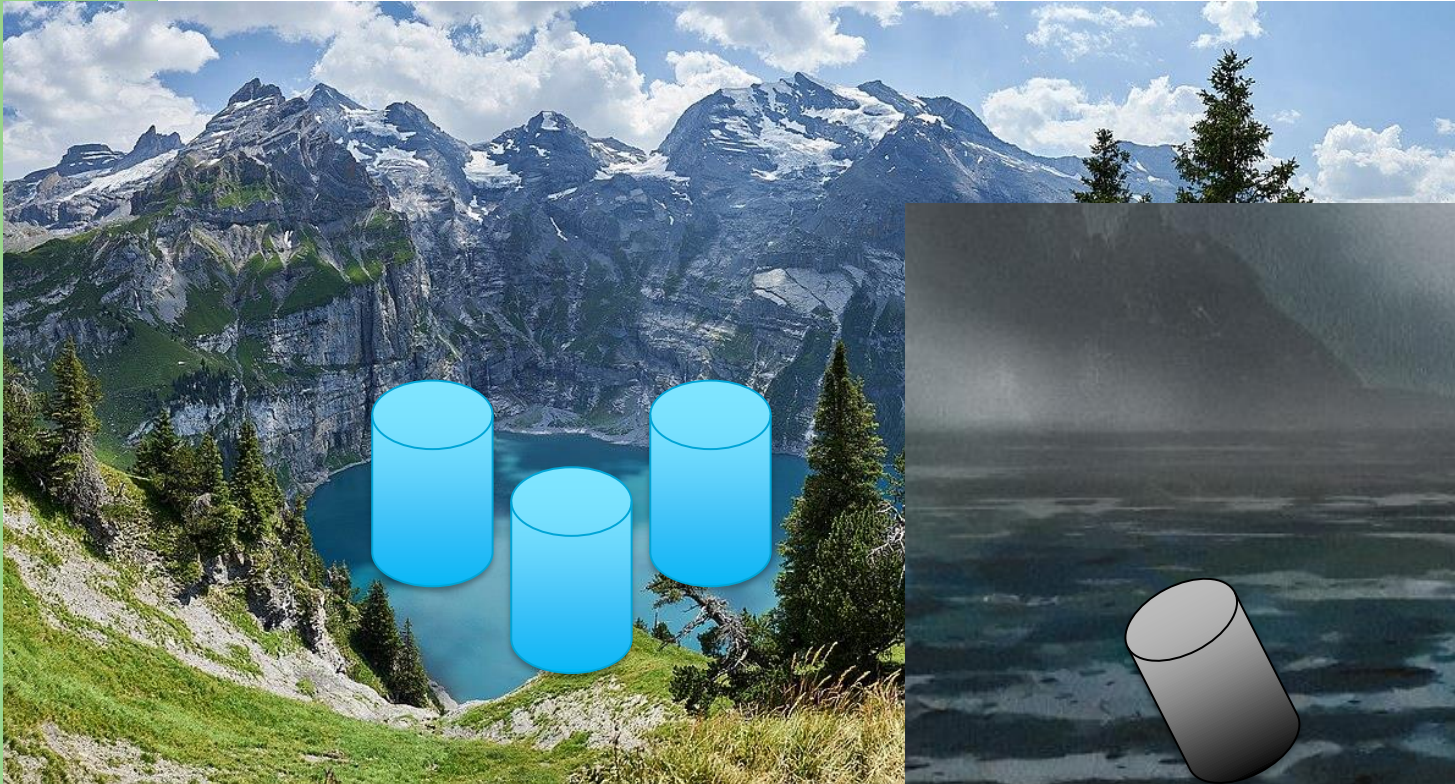
Semantics-Based Data Engineering

- Big Variety
 - Source Data: Unstructured implicit semantics
 - Target Data: Structured explicit semantics
- Big Veracity
 - Can we trust our input data?
 - Is it useful for the task?

Data Lakes?



Data Lakes? Or Data Swamp?



Big Variety – Data Integration

- The core challenge for Big Data Systems:
 - Data INTEGRATION
 - “Big Variety is the 800-pound gorilla in the corner.”
 - Michael Stonebreaker, during Touring Awards 2015
 - Data CLEANING
 - “The 80/20 data science dilemma”

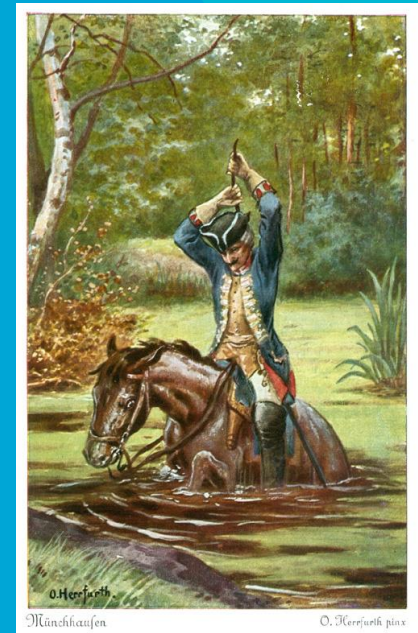


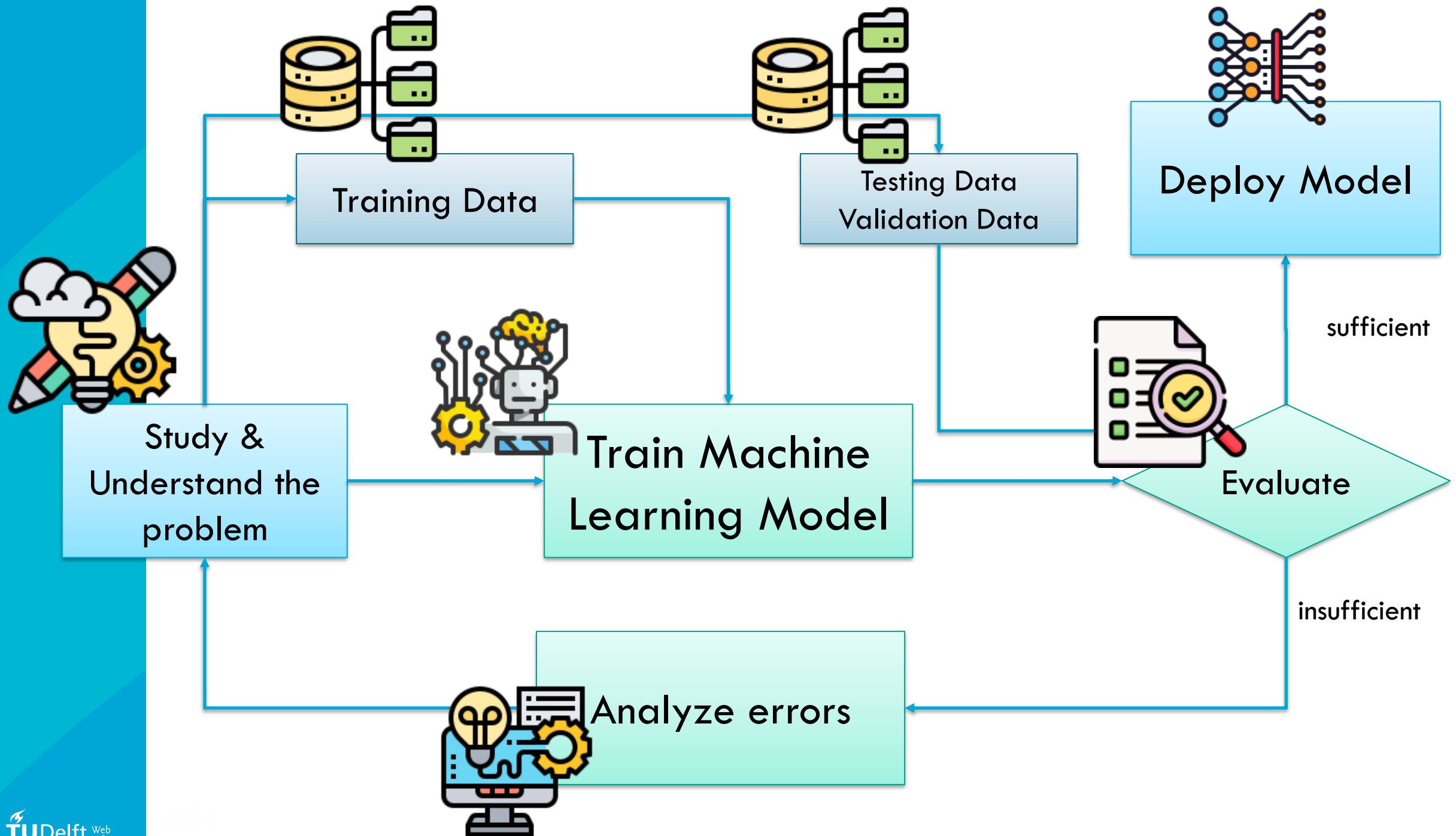
Big Veracity

- Core challenge here:
 - Is my data correct?
 - Is my data good?
 - Is my data useful for the purpose?
 - Can I actually answer any of the above questions?

Where does all my data come from?

Make it, buy it, steal it, fabricate it...





Where does ML data come from?

1. Reuse Open ML Training Data
 - Platforms like Kaggle, or your university course providers
2. Data Repositories and Data Marts
 - “Buy” or “Get” data
3. Your own Enterprise Database Management System
 - “Make your own data”
4. Data Farms / Crowd Sourcing
 - Use (and pay?) a crowd of humans to make the data for you
5. Artificial/Synthetic Data Training Data
 - “Use AI-generated data to train another AI”
6. “Steal It”
 - Crawl data from everywhere without any consideration about rights and wrong whatsoever....

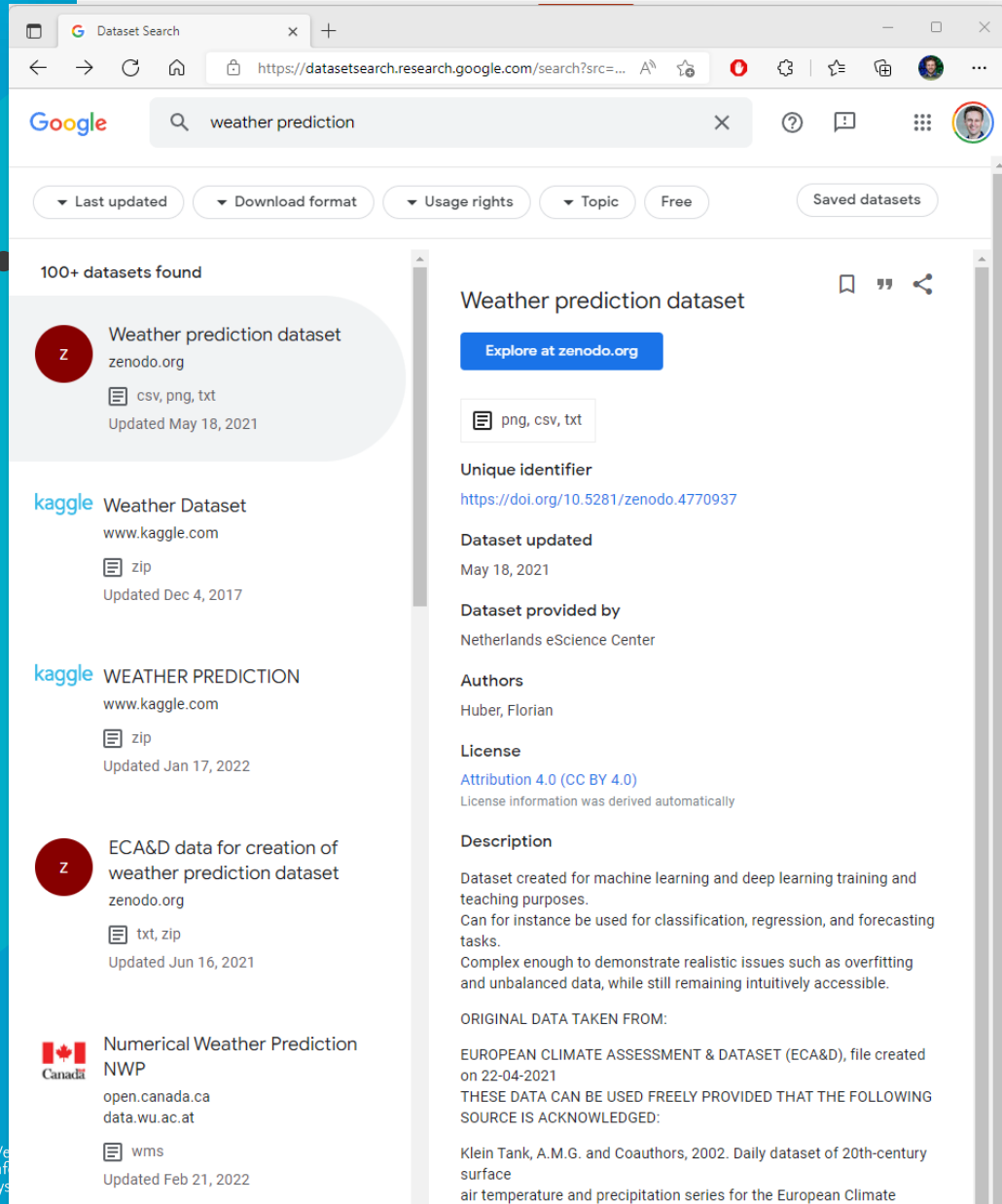
1 Reuse Open ML Training Data

- Platforms like Kaggle, or your university course providers
- These datasets are typically **taylor-made** for a specific machine learning task!
 - Are they **relevant** to YOUR problem?

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 - Are they **relevant** to YOUR problem?

1 Reuse Open ML Training



Dataset Search results for "weather prediction". The search shows 100+ datasets found. The top results are:

- Weather prediction dataset** by zenodo.org. Formats: csv, png, txt. Updated May 18, 2021. [Explore at zenodo.org](#)
- Weather Dataset** by kaggle.com. Format: zip. Updated Dec 4, 2017.
- WEATHER PREDICTION** by kaggle.com. Format: zip. Updated Jan 17, 2022.
- ECA&D data for creation of weather prediction dataset** by zenodo.org. Formats: txt, zip. Updated Jun 16, 2021.
- Numerical Weather Prediction NWP** by open.canada.ca. Format: wms. Updated Feb 21, 2022.

The detailed view of the "Weather prediction dataset" shows it is provided by the Netherlands eScience Center, has a Unique identifier of <https://doi.org/10.5281/zenodo.4770937>, and is licensed under Attribution 4.0 (CC BY 4.0).

or unive
tailor-

problem



Registry of Open Data on AWS. The page lists various open datasets available on AWS. The featured dataset is "The Cancer Genome Atlas (TCGA)", a collaboration between the National Cancer Institute (NCI) and National Human Genome Research Institute (NHGRI). It aims to generate comprehensive, multi-dimensional maps of the key genomic changes in major types and subtypes of cancer. The dataset contains open Clinical Supplement, Biospecimen Supplement, RNA-Seq Gene Expression Quantification, miRNA-Seq Isoform Expression Quantification...

Use examples

- ISB Cancer Genomics Cloud by Institute for Systems Biology
- Comprehensive Characterization of Cancer Driver Genes and Mutations by Matthew H. Bailey, Collin Tokheim, et al.
- An Integrated TCGA Pan-Cancer Clinical Data Resource to Drive High-Quality Survival Outcome Analytics by Jianfang Liu, Tara Lichtenberg, et al.
- The Immune Landscape of Cancer by Vésteinn Thorsson, David L. Gibbs, et al.
- Broad Institute FireCloud by The Broad Institute of MIT & Harvard

See 29 usage examples →

Foldingathome COVID-19 Datasets

Folding@home is a massively distributed computing project that uses biomolecular simulations to investigate the molecular origins of disease and accelerate the discovery of new therapies. Run by the Folding@home Consortium, a worldwide network of research laboratories focusing on a variety of different diseases, Folding@home seeks to address problems in human health on a scale that is infeasible by any other means, sharing the results of these large-scale studies with the research community through peer-reviewed publications and publicly shared datasets. During the COVID-19 epidemic, Folding@home focused its resources on understanding the vulnerabilities in SARS-CoV-2, the virus that causes COVID-19 disease, and working closely with a number of experimental collaborators to accelerate progress toward effective therapies for treating COVID-19 and ending the pandemic. In the process, it created the world's first exascale distributed computing resource, enabling it to generate valuable scientific datasets of unprecedented size. More information about Folding@home's COVID-19 research activities at the Folding@home COVID-19 page. In addition to working directly with experimental collaborators and rapidly sharing new research findings through preprint servers, Folding@home has joined other researchers in committing to rapidly share all COVID-19 research data, and has joined other researchers in committing to rapidly share all COVID-19 research data, and has joined other researchers in committing to rapidly share all COVID-19 research data.

2. Data Repositories and Data Marts

- “Buy” or “Get” data
- Many companies and institutions collect large datasets and provide them for a fee (or sometimes for free)
- Often, these datasets are not made specifically for ML training, and to augmented before use (like adding **labels**)

StatLine

Thema'sRecentHelp

Zoeken...



Nederland in cijfers

[Naar thema's](#)

StatLine

StatLine is de databank van het CBS. Het CBS biedt een schat aan cijfers over de Nederlandse economie en samenleving. Van inflatie tot bevolkingsontwikkeling, de informatie is overzichtelijk gerubriceerd op thema en gratis beschikbaar voor iedereen.

[Tabellen naar thema](#)

Zoek op trefwoord

U kunt de data met trefwoorden doorzoeken en krijgt dan een lijst met tabellen waar uw zoekterm in voorkomt. Deze lijst kunt u vervolgens verfijnen aan de hand van thema, regionale indeling of verslagjaar.

[Direct zoeken](#)

Grafieken en kaarten

Het is mogelijk met StatLine om met een druk op de knop grafieken en kaarten maken. Aanpassen van en switchen tussen grafieken is ook mogelijk. Dit is met handige iconen eenvoudig te doen. De CBS-cijfers zijn hierdoor eenvoudig visueel te presenteren en te delen.

Uitleg

Hoe zit StatLine in elkaar en wat kunt u er allemaal wel en niet mee?

Voor optimaal gebruiksgemak:

[Handleiding en demo-filmpjes](#)
[Veelgestelde vragen](#)

Datum	Publicatie
13-07-2022	Luchtvaart: vliegtuigbewegingen op kleine luchthavens
12-07-2022	Faillissementen: natuurlijke personen, leeftijd
12-07-2022	Faillissementen: natuurlijke personen, regio
12-07-2022	Faillissementen, zittingsdaggecorrigeerd
12-07-2022	Faillissementen: kerncijfers

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Overheid.nl

Open data of the government

Overheid.nl Home Data Impact Community News Support Publish a dataset

Dataset

What are you looking for? Search

Filter your results

Selected filters

Theme: Education and science X

Publicity level: Public X

Clear all filters

578 datasets

Sort by: Relevance / Last modified / Popularity

State (1)

Available (578)

Data owner (22)

Level of governance (5)

Rijk (538)

Gemeente (31)

Overig (7)

Provincie (1)

Regionale samenwerking (1)

Theme (16)

Classification (3)

Landelijke dekking (401)

High value (7)

Referentiedata (2)

Group (11)

License (4)

DATASET

VSV; totaalcijfer OCW (jongeren tot 23 jaar)

Deze tabel geeft informatie over het aantal voortijdig schoolverlaters (jaarlijkse aanwas) tot 23 jaar vanuit het voortgezet onderwijs (vo) en het middelbaar beroepsonderwijs (mbo). Let op: In deze tab...

Data owner:

Centraal Bureau voor de Statistiek (Rijk)

Updated:

06/17/2022 - 00:00

Theme:

Education and science

Description Download Webservice Metadata

DATASET

Arbeidsmarktpositie van pas afgestudeerden

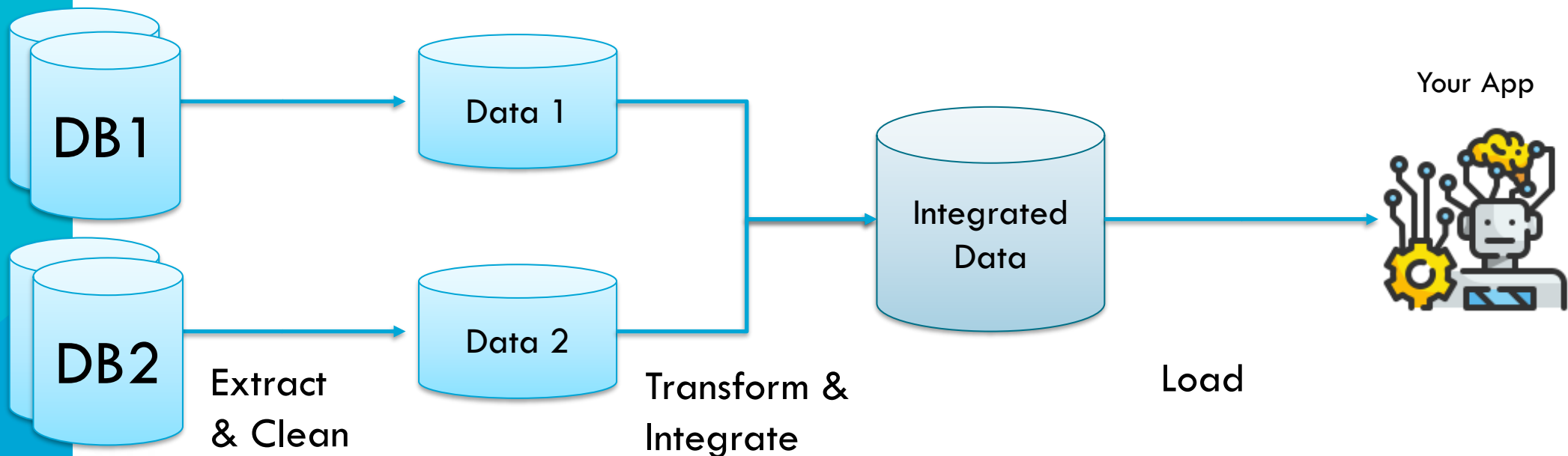
Arbeidsmarktpositie (incl. startsalaris) van pas afgestudeerde hbo'ers en academici naar product...

3. Your own Enterprise Database Management System

- “Make your own data”
- You might already own data yourself in your enterprise/app!
 - If you run a web shop, you already have data on customers, items, and purchases
 - If you run a community site, you already have data on user content and user interactions
 - How **extract** that data from your database systems?
- Also, is this really your data? Are you allowed to use it?
 - See **General Data Protection Regulation (GDPR)** or **European Data Act**

3. Your own Enterprise Database Management System

- We often talk a about **Data Pipelines** in this context
- Or **ETL**: Extract, Transform, Load



4. Data Farms / Crowdsourcing

- Especially, **labels** are very hard to come by
 - Many labeling tasks require a real (intelligent?) person
 - e.g., detect complex objects in images, detect hate speech, detect irony, annotate emotions, etc.
- Coordinate many people to make labels /data for you
 - ...recruiting paid or unpaid volunteers > Crowdsourcing
 - ...as a centralized large-scale business > Data Farm

4. Data Farms / Crowdsourcing

- ...”recruiting” paid or unpaid volunteers
Crowdsourcing

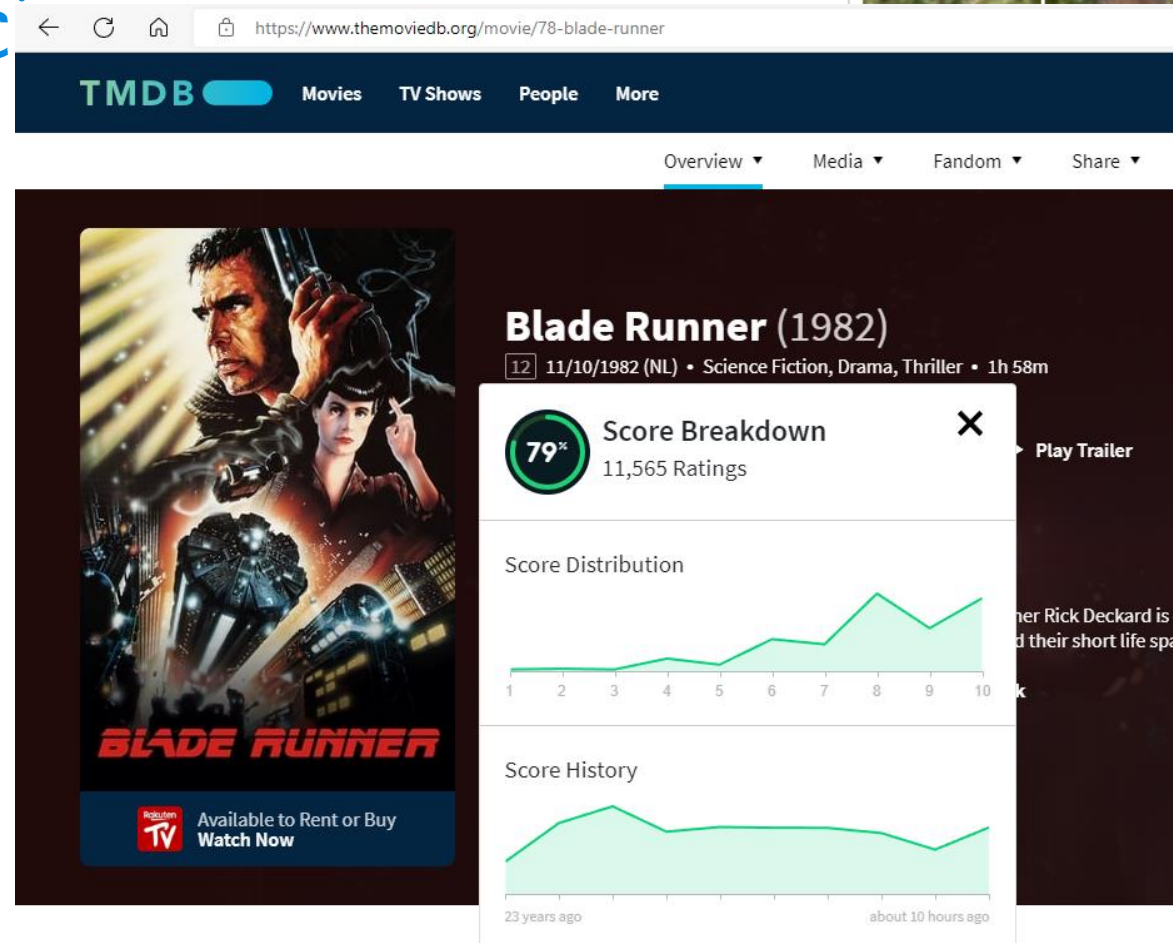


Piggybacking

From Google reCAPTCHA: <https://www.google.com/recaptcha/>

4. Data Farms / Crowdsourcing

- ...”recruiting” paid or unpaid volunteers > Crowdsourcing



<https://www.google.com/recaptcha/>

4. Data Farms / Crowdsourcing

Amazon Mechanical Turk

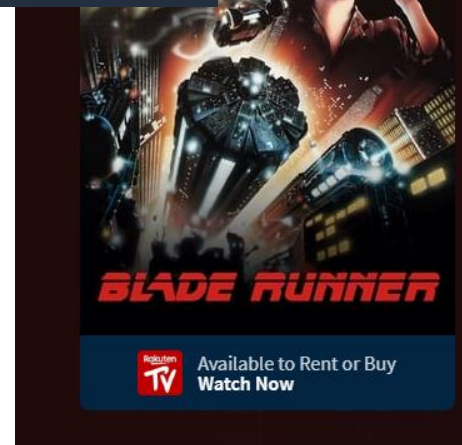
Access a global, on-demand, 24x7 workforce



Paid Microtask Crowdsourcing

from Amazon Mechanical Turk: <https://requester.mturk.com/>

paid or unpaid volunteers >



Blade Runner (1982)

12 11/10/1982 (NL) • Science Fiction, Drama, Thriller • 1h 58m



Score Distribution



Score History



<https://www.google.com/recaptcha/>

Community Contributions from TheMovieDB: <https://www.themoviedb.org/movie/78-blade-runner>

4. Data Forms / Crowdsourcing

Amazon Mechanical Turk

Access a global



Paid Microtask Crowdsourcing
from Amazon Mechanical Turk: <https://www.mturk.com/>

National Bee Count

Working together to protect wild bees

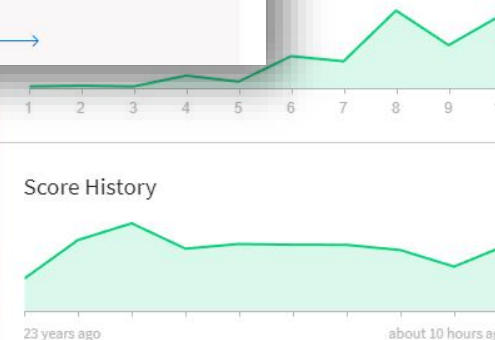
Every year, people in the Netherlands are urged to count bees as part of the National Bee Count. By spending just half an hour counting all of the bees in your garden, on your balcony or some other green area in your neighborhood, you can help us to protect wild bees. The data on the number of bee species observed in the country is vital to scientists seeking to learn more about the state of play in the wild bee population. The more they learn, the better they are able to protect this immensely useful and vulnerable species of animal.



Get counting!

It's easy to take part, even if you have little to no knowledge of bees. You can find a count form and a bee guide that allows anyone to recognize the sixteen different types of bees, bumblebees and hoverflies. Visit the National Bee Count website to see when the next count will be taking place and get involved!

[Get counting \(Dutch\)](#) →



Select squares with
street signs
If there are none, click skip

volunteers >



SKIP

<https://www.google.com/recaptcha/>

Community Contributions from TheMovieDB: <https://www.themoviedb.org/movie/78-blade-runner>

4. Data Farms / Crowdsourcing

Technology Quarterly
Jan 2nd 2020 edition >

China's success at AI has relied on good data

But cheap labour has also played an important part



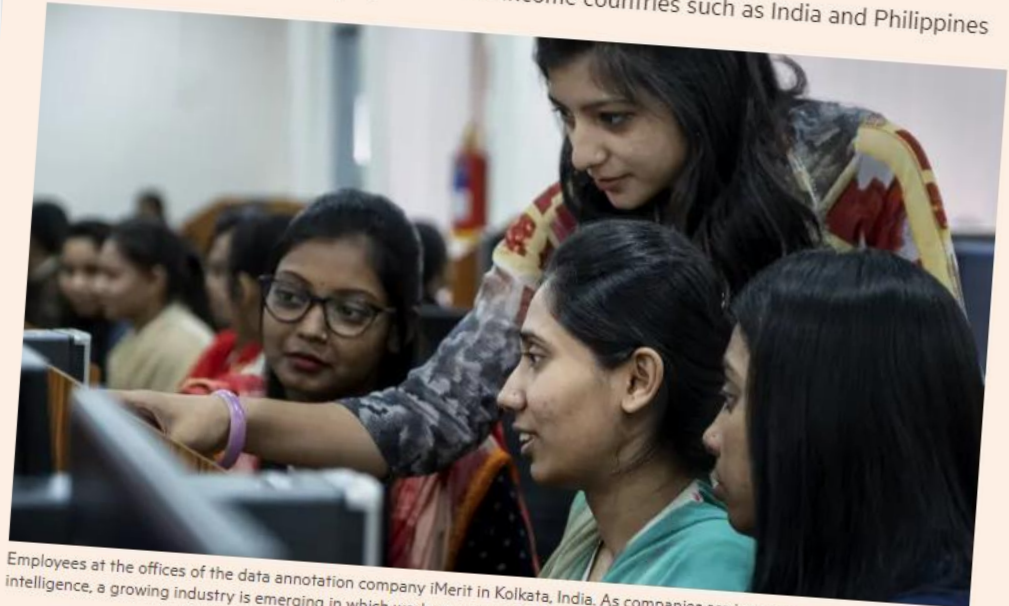
CHINA IS THE land of face recognition. Cameras able to extract face prints from passers-by are common in the streets of large cities like Guangzhou and Shenzhen. Boxy vending machines at airports offer to let you pay for a cup of orange juice, robot-squeezed for perfect freshness, by scanning your face. From December 1st all people applying for an account with one of China's telecoms companies such as China Mobile must have their face scanned. Previous regulations required proof of identity, but the possession of users' face prints will let firms verify identities in real-time via smartphone cameras.

Considering the oppressive purposes to which this technology is being put—most notably in the Muslim-majority areas of north-west China—it would not be appropriate to call China's rapid adoption of it anything more than a technical success. The underappreciated fact that companies leaping ahead in the field are using cheap labour for their progress than on any other factor before declaring a

Artificial intelligence + Add to myFT

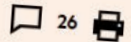
AI's new workforce: the data-labelling industry spreads globally

Hundreds of thousands employed in lower-income countries such as India and Philippines



Employees at the offices of the data annotation company iMerit in Kolkata, India. As companies are increasingly embracing artificial intelligence, a growing industry is emerging in which workers are employed to 'train' algorithms to recognise various types of data

Madhumita Murgia JULY 24 2019



On the fringes of the Indian city of Kolkata, in the dusty, crowded neighbourhood of Metiabruz, 460 young women are working at the vanguard of artificial intelligence.

The women, mostly from the local Muslim community, are helping to train computer vision algorithms used in autonomous vehicles and augmented reality systems, for the likes of Amazon, Microsoft, eBay and TripAdvisor.

4. Data I

Technology Quarterly
Jan 2nd 2020 edition >

Data

China's on goo

But cheap labor



CHINA from and Sher orange j Decemb compar regulati will let

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BUSINESS

The New York Times

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THE NEW NEW WORLD

How Cheap Labor Drives China's A.I. Ambitions

Give this article

17



Workers at the headquarters of Ruijin Technology Company in Jiaxian, in central China's Henan Province. They identify objects in images to help artificial intelligence make sense of the world. Yan Cong for The New York Times

By **Li Yuan**

Nov. 25, 2018

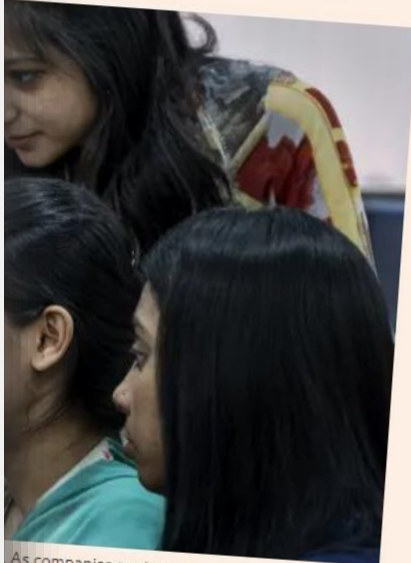
阅读简体中文版

閱讀繁體中文版

Some of the most critical work in advancing China's technology goals takes place in a former cement factory in the middle of the

Data-labelling industry

countries such as India and Philippines



As companies are increasingly embracing artificial 'ain' algorithms to recognise various types of data

26

dusty, crowded
re working at the vanguard

nity, are helping to train
ehicles and augmented
eBay and TripAd

Technology Quarterly
Jan 2nd 2020 edition >

China's on goo
But cheap labor



China's on goo
But cheap labor

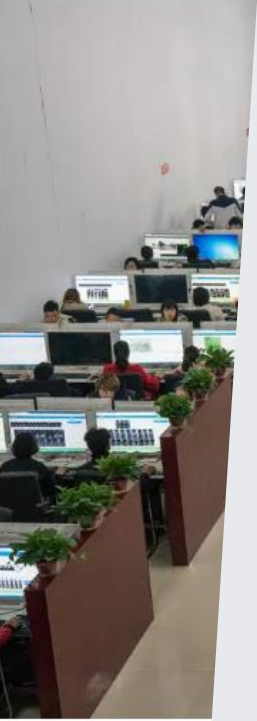
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THE NEW

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Workers at the headquarter
They identify objects in ima
York Times

THE VERGE

THE TRAUMA FLOOR

The secret lives of Facebook moderators in America

By Casey Newton | @CaseyNewton | Feb 25, 2019, 8:00am EST
Illustrations by Corey Brickley | Photography by Jessica Chou

If you buy something from a Verge link, Vox Media may earn a commission. See our [ethics statement](#).



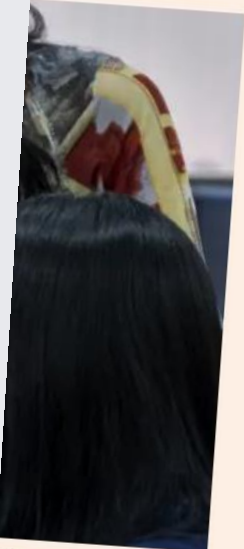
Content warning: This story contains discussion of serious mental health issues and racism.

The panic attacks started after Chloe watched a man die.

She spent the past three and a half weeks in training, trying to harden herself against the daily onslaught of disturbing content.

g industry

dia and Philippines



reasingly embracing artificial
gnise various types of data

26

led
at the vanguard

ping to train
augmented
ria A. d.

But what if you never had the data to start with?

- From the Economist, and Cognilytica:
 - 3rd party data preparation market was 1.5bn USD in 2019, expected 3.5bn USD in 2024
 - 3rd party labeling 1.7bn USD 2019, 4.1bn USD in 2024



5. Artificial / Synthetic Data

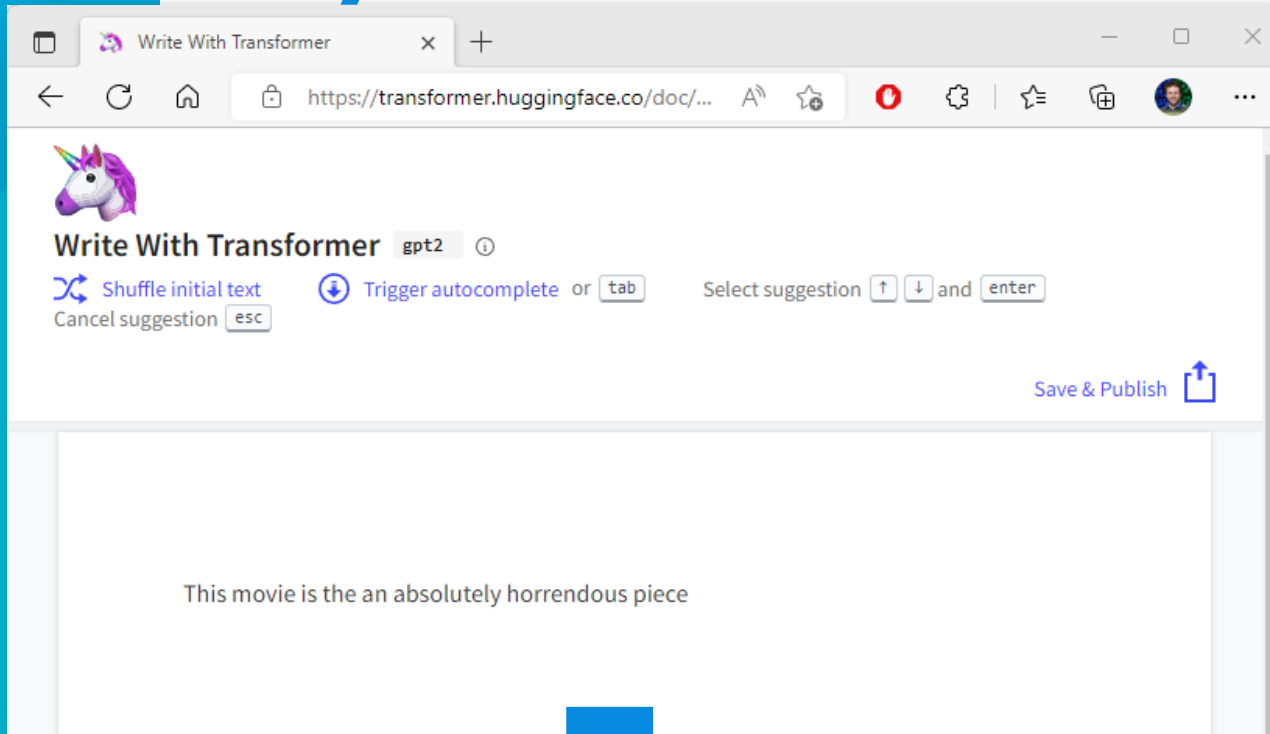
- Hot academic topic!
- Core Idea: Train an AI to learn a **representation** of your data, and then use that to generate more like it
 - For example, using **Variational Autoencoders**
- Such data is not “**real**”, and thus “**lower quality**”
 - ...but you can make a lot of it, and often volume beats quality!
- Works well for **augmenting** real datasets
 - e.g., balance under-represented classes

Synthetic Text

from Huggingface / GPT2:

<https://transformer.huggingface.co/doc/gpt2-large>

<https://openai.com/blog/tags/gpt-2/>



This movie is the an absolutely horrendous piece of filmmaking, one that is filled with absolutely no story, no characters, and really no sense of purpose. It just tries to use the story of what's going on right now to sell its premise. This is the worst possible movie ever made and I can't really think of a better way to spend a Saturday night than watching it.

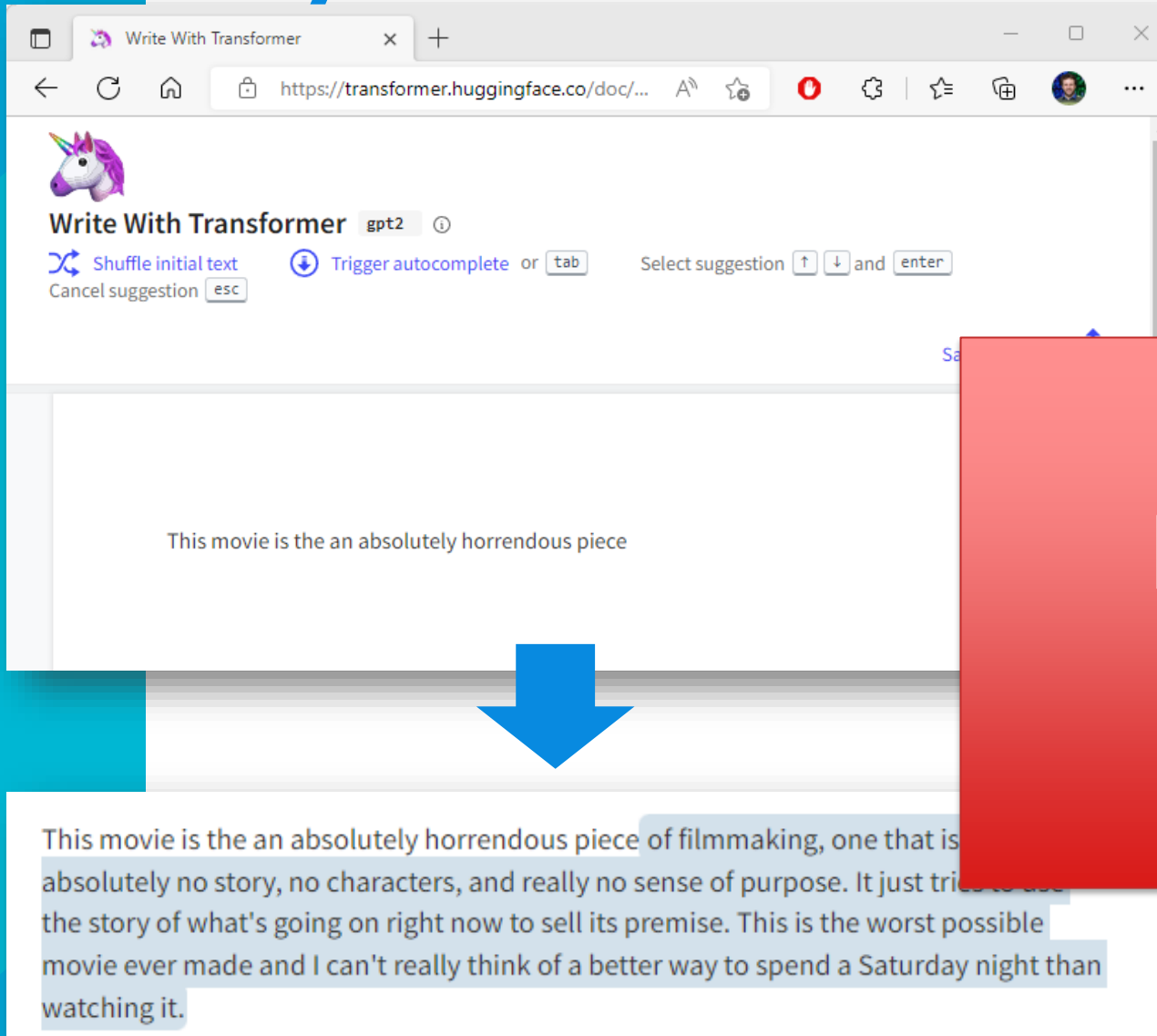
???

Synthetic Text

from Huggingface / GPT2:

<https://transformer.huggingface.co/doc/gpt2-large>

<https://openai.com/blog/tags/gpt-2/>



HAHA! Slides from
2022...time flies!!!

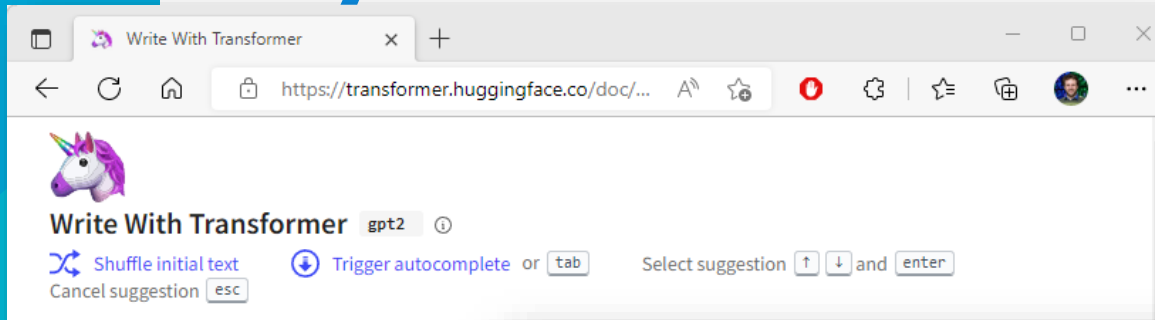
???

Synthetic Text

from Huggingface / GPT2:

<https://transformer.huggingface.co/doc/gpt2-large>

<https://openai.com/blog/tags/gpt-2/>



!!??!!!

The Netherlands is a hotspot for international drug trafficking and has the world's highest murder rate, with one in five residents said to be involved. The most dangerous city in the world to live in, Amsterdam, is home to more than a quarter of the world's meth labs. In a study of the most common drugs sold in the Netherlands, researchers found that the most frequently bought and sold drugs in the Netherlands are cocaine , ecstasy, ketamine and heroin. Most of the cocaine and ecstasy sold in the Netherlands are bought by people aged 30 or over , with the average Dutch adult buying at least six drugs every month.

Written by Transformer · transformer.huggingface.co 🦄

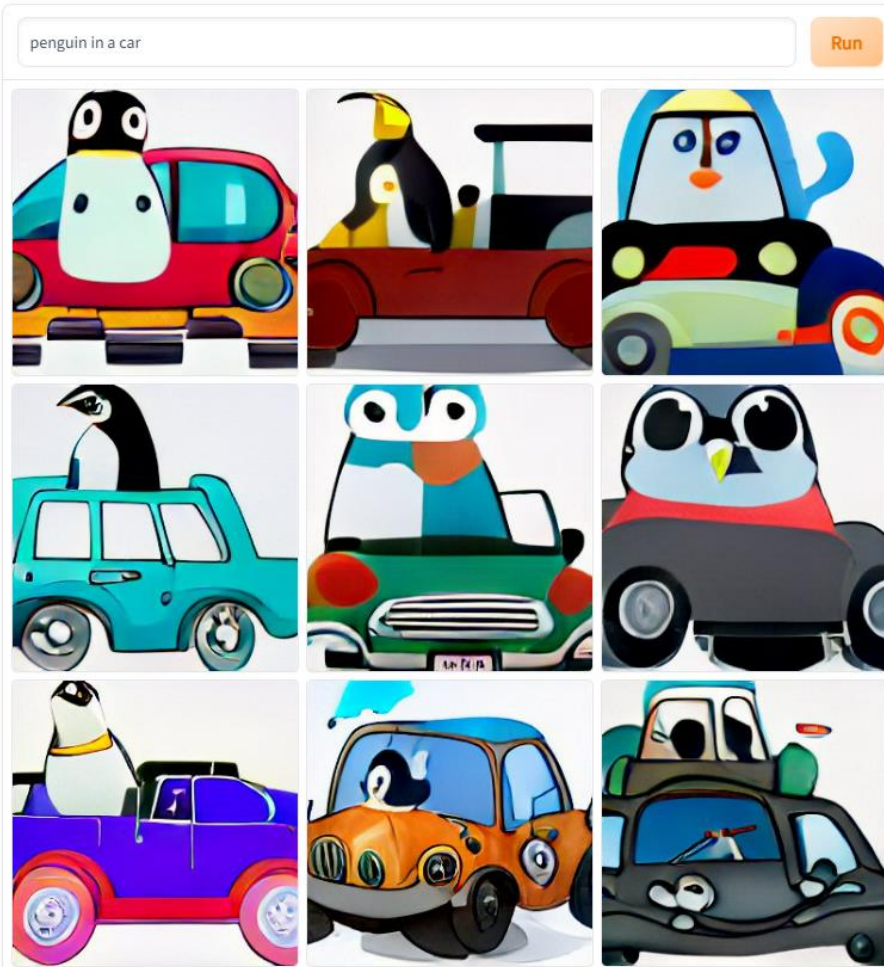
This movie is the an absolutely horrendous piece of filmmaking, one that is filled with absolutely no story, no characters, and really no sense of purpose. It just tries to use the story of what's going on right now to sell its premise. This is the worst possible movie ever made and I can't really think of a better way to spend a Saturday night than watching it.

Synthetic Images

“Penguin in a car”

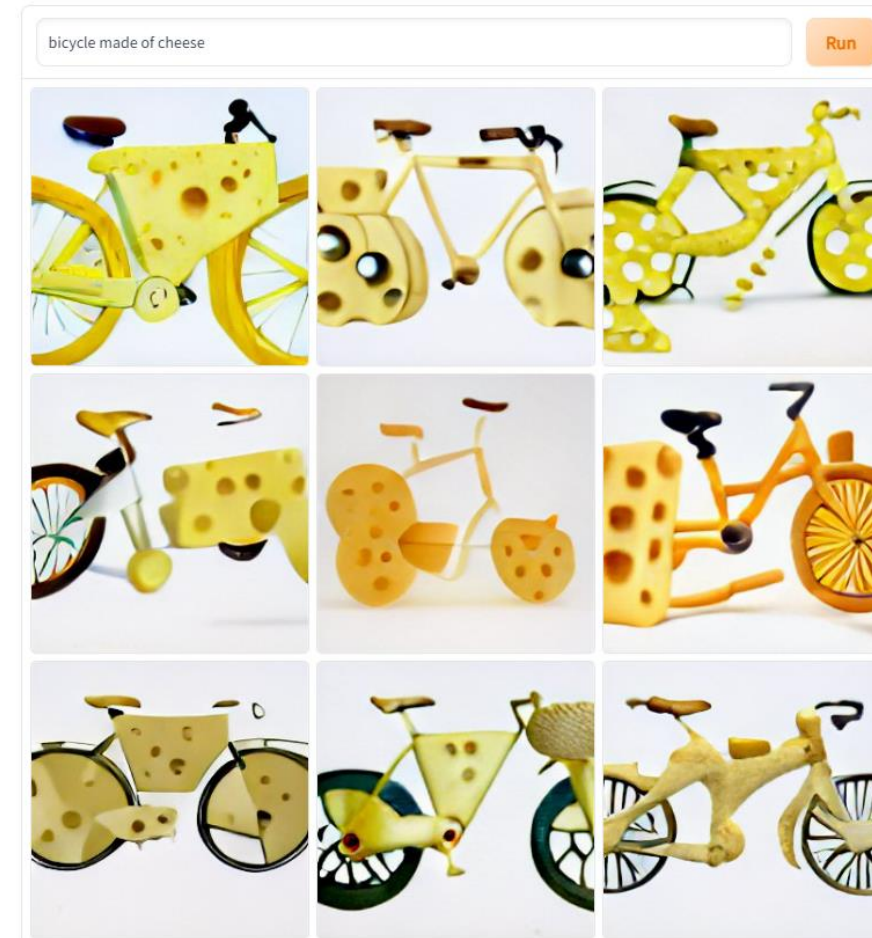
DALL·E mini

AI model generating images from any prompt!



from Huggingface / OpenAI:
[DALL·E mini - by craiyon.com \(huggingface.co\)](https://huggingface.co/craiyon/dall-e-mini)
<https://openai.com/dall-e-2/>

“Bicycle made of cheese”



- 2024 😊



Synthetic Images

from: Ramesh, Aditya, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. "Zero-shot text-to-image generation." In *International Conference on Machine Learning*, pp. 8821-8831. PMLR, 2021.

Also: <https://cv-tricks.com/how-to/dall-e-text-to-image-generation/>



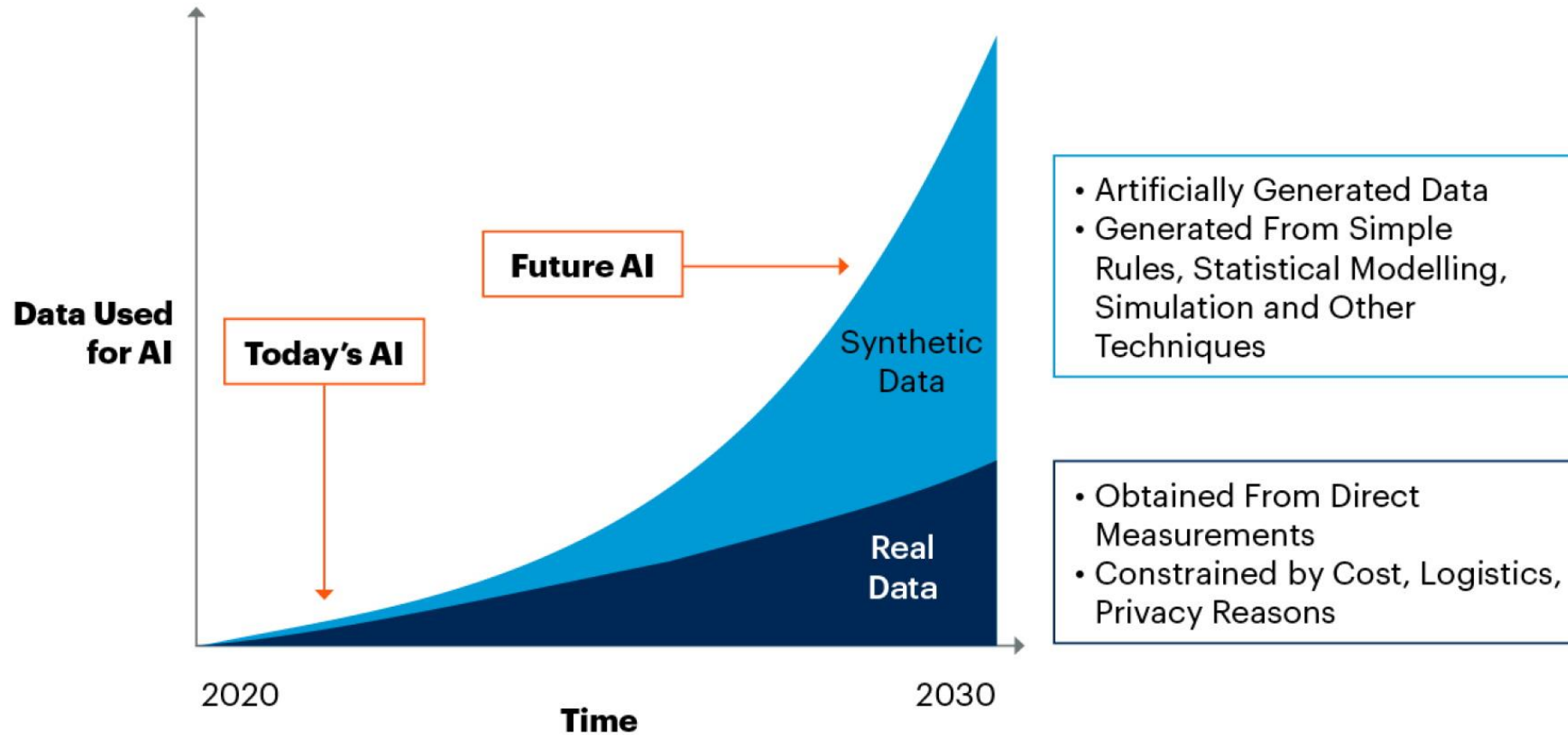
"A road sign with an image of a blue strawberry"

Synthetic Images 2024 Edition

A road next to Institut für Informatik in Zürich / Oerlikon clearly showing a traffic sign with a blue strawberry on it. the sign has a circle shape, and the strawberry is heavily stylized and symbolized.



By 2030, Synthetic Data Will Completely Overshadow Real Data in AI Models



Source: Gartner
750175_C

Gartner

From:

Synthetic data will become the main form of data used in AI. Source: Gartner, "Maverick Research: Forget About Your Real Data – Synthetic Data Is the Future of AI," Leinar Ramos, Jitendra Subramanyam, 24 June 2021

6. “Steal it”...or obtain in other dubious ways

LAUREN GOODE BUSINESS SEP 25, 2023 6:47 PM

Getty Images Plunges Into the Generative AI Pool

Stock-photo giant Getty Images has partnered with Nvidia to build an image generator. Just like with other tools of its ilk, questions remain about who should get credit for the pictures it dreams up.

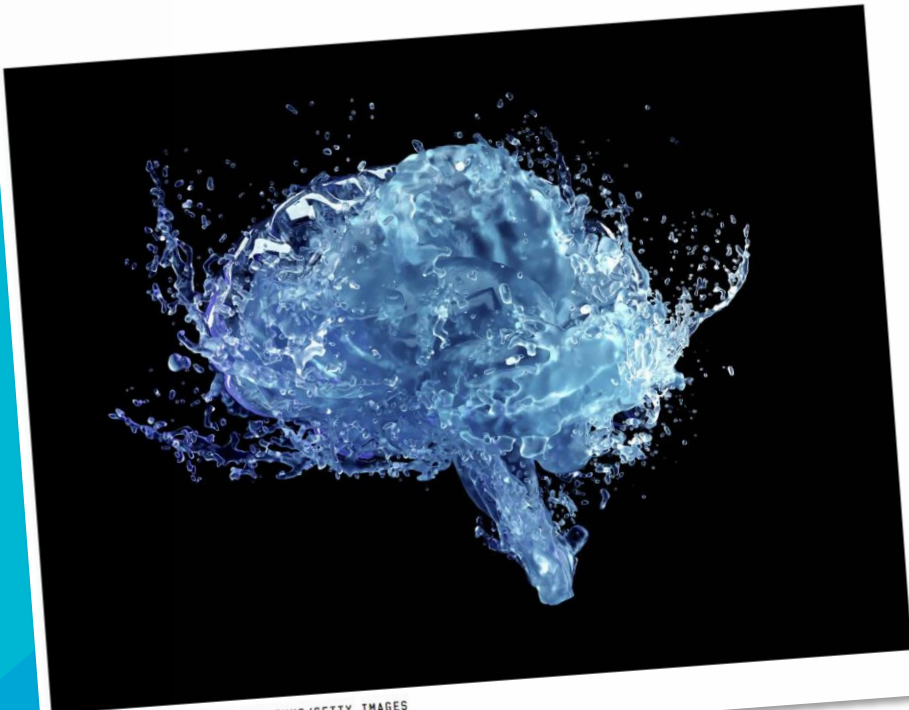


ILLUSTRATION: ANDRIY DNUFRIYENKO/GETTY IMAGES

6. “Steal it”...or obtain in other dubious ways

LAUREN GOODE BUSINESS SEP 25, 2023 6:47 PM

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ILLUSTRATION: ANDRIY DNUFRIYENKO/GETTY IMAGES



6. “Steal it”...or obtain in other dubious ways

ChatGPT Stole Your Work. So What Are You Going to Do?

Creators need to pressure the courts, the market, and regulators before it's too late.

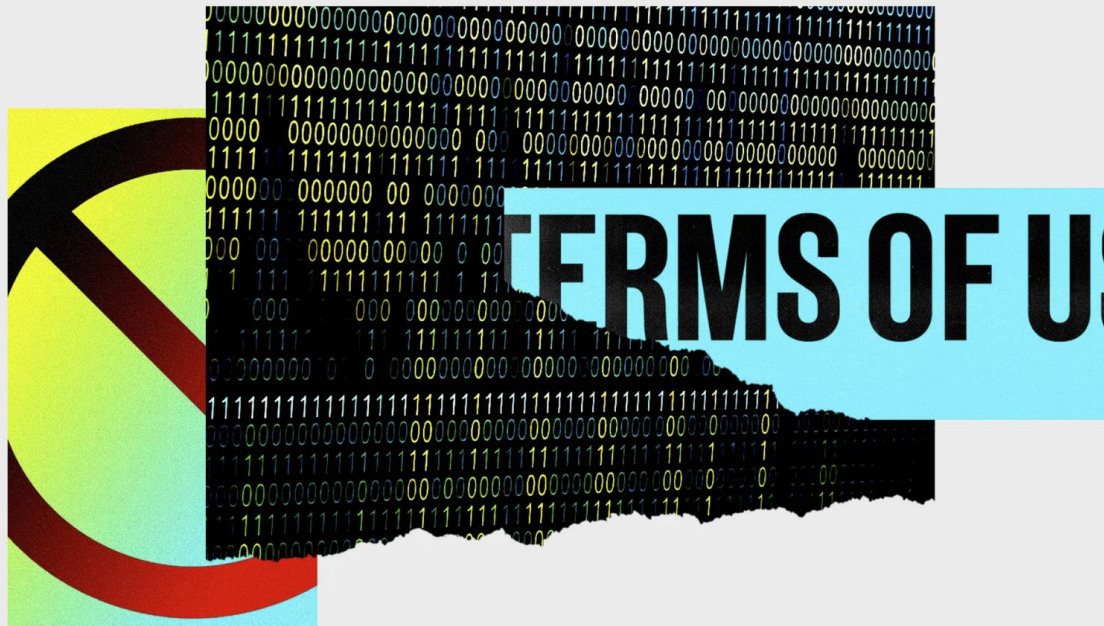


PHOTO-ILLUSTRATION: WIRED STAFF; GETTY IMAGES

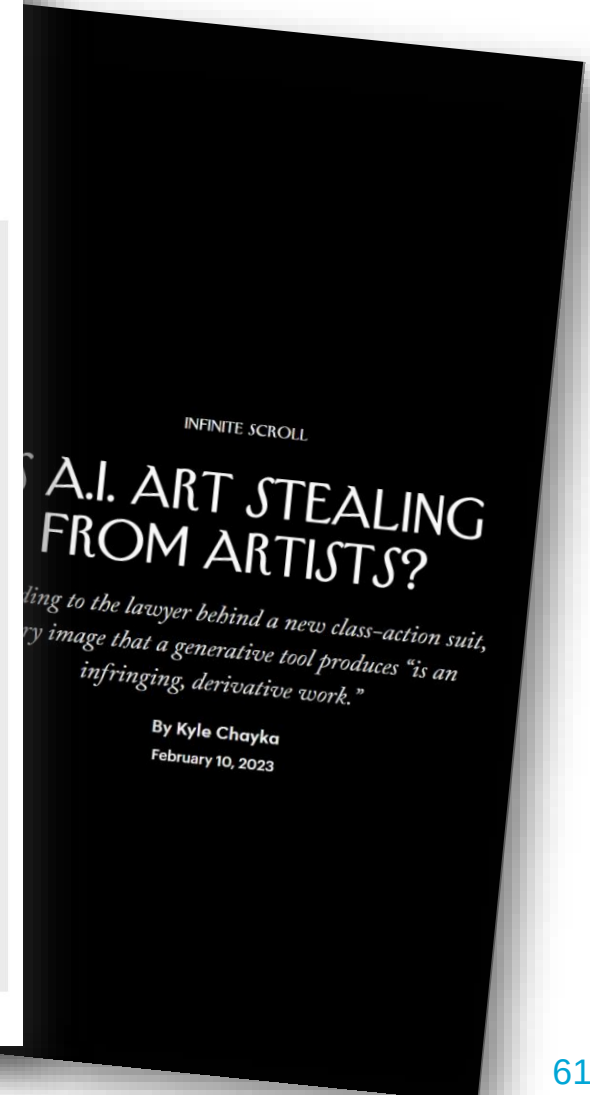
LAUREN GOODE BUSINESS SEP 25, 2023 6:47 PM

Getty Images Plunge

Stock-photo giant Getty Images has partner questions remain about who should get credit



ILLUSTRATION: ANDRIY DNUFRIYENKO/GETTY IMAGES



6. "Steal it"

ENTERTAINMENT > MOVIES

<https://people.com/sag-claims-studios-want-to-scan-background-actors-use-likenesses-for-free-7561285>

Hollywood Studios Want to Scan Background Actors for One Day's Pay, Use Likenesses for 'Rest of Eternity,' SAG Claims

In a statement amid the strike decision, the AMPTP said that their offer included "a groundbreaking AI proposal that protects actors' digital likenesses"

By **Jen Juneau** | Published on July 14, 2023 03:26PM EDT



PHOTO: JOHN SALANGSANG/SHUTTERSTOCK

6. “Steal it”...or obtain in other dubious ways

- We don't really know yet what to do here
 - Legal aspects?
 - What is allowed and forbidden?
 - How do we deal with this internationally???
 - Ethical aspects?
 - Just because it's legal doesn't mean that it's right

Is my AI system a sexist Nazi?

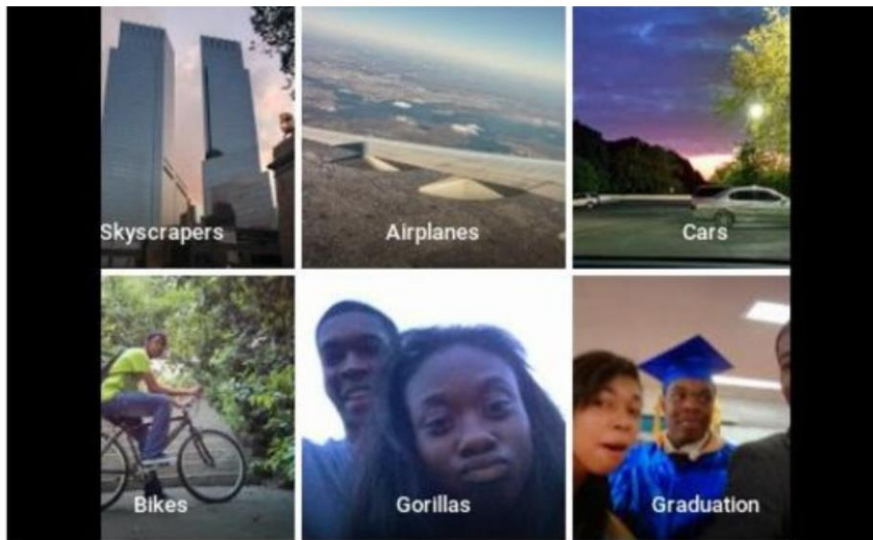
...or did it just feed on bad data?

Bias

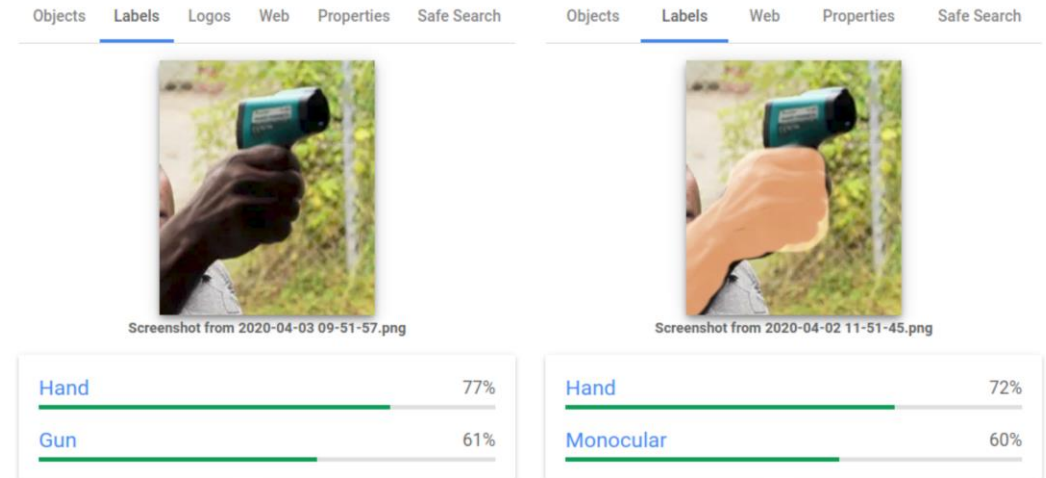
Google apologises for Photos app's racist blunder

1 July 2015

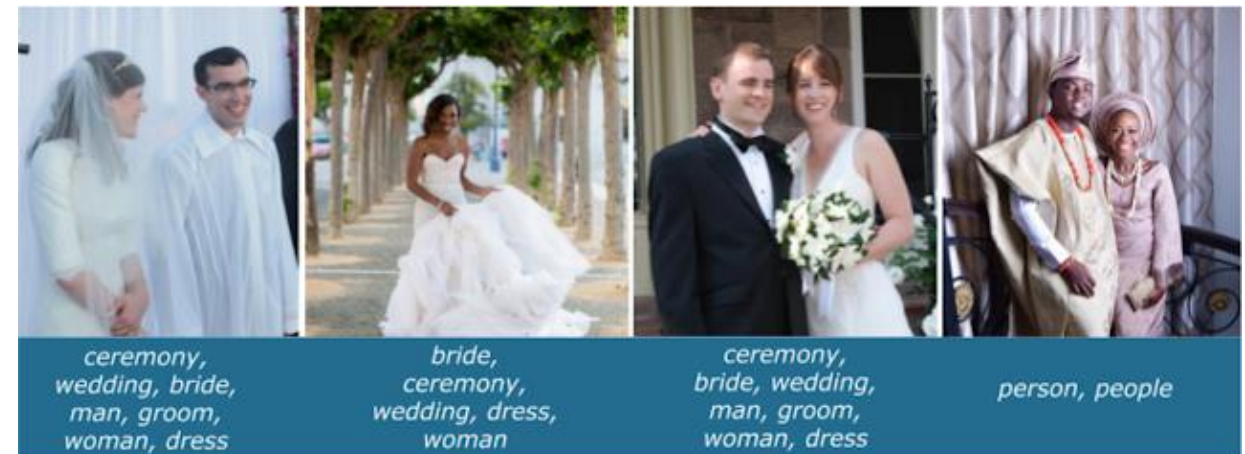
f     Share



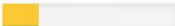
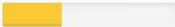
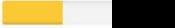
diri noir avec banan @jackyalcine · Jun 29
Google Photos, y'all [redacted] My friend's not a gorilla.

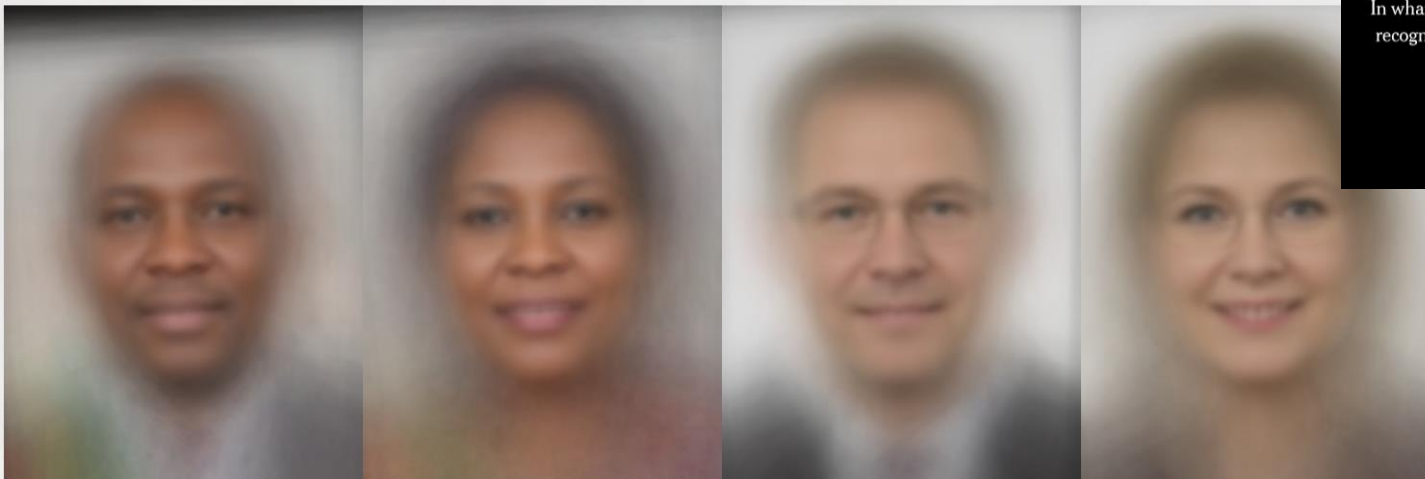
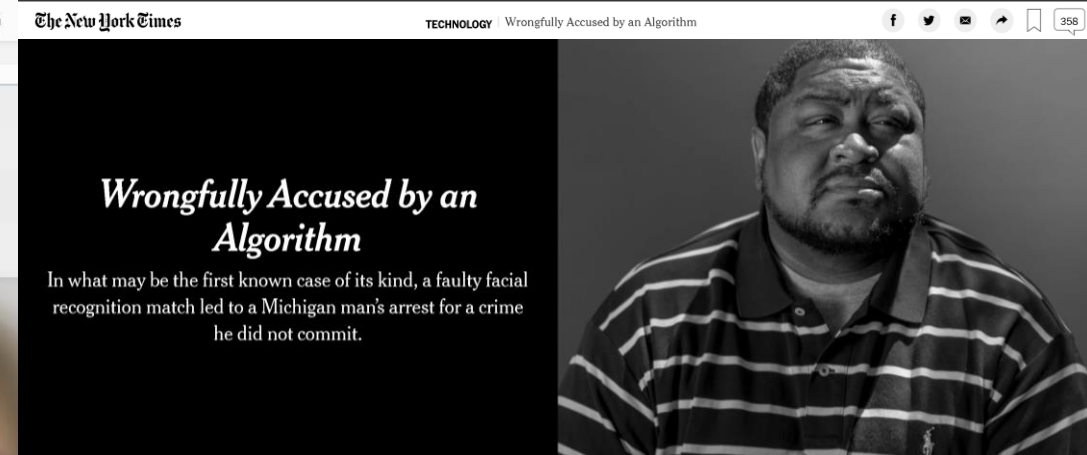


On 3 April, Google Vision Cloud produced starkly different labels after an overlay was added.



Large training datasets of possibly low diversity
Deep learning model learned wrong prediction features

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8% 
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8% 
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4% 



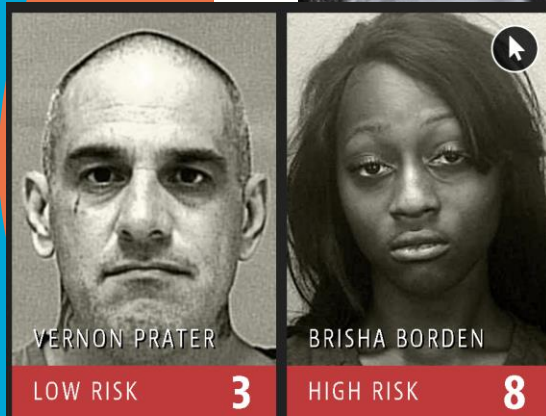
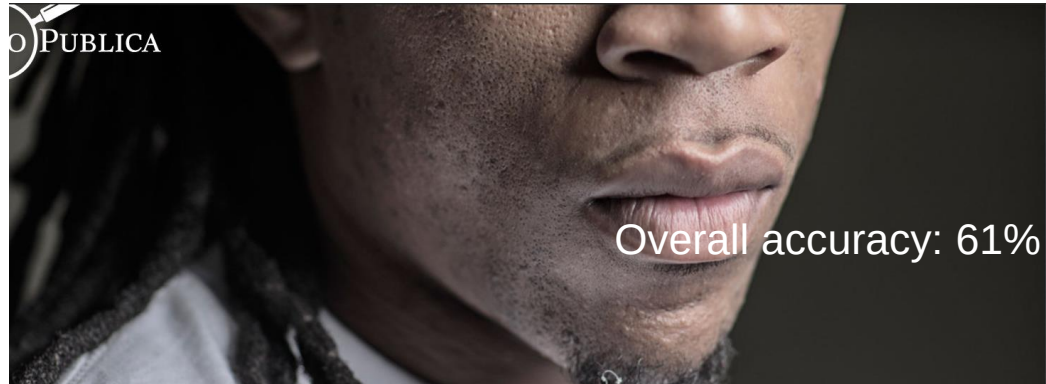
Unbalance in training data for different populations
→ Lack of requirements engineering

Attributes:

- defendant's demographic information (e.g., gender, race),
- criminal history (e.g., number of prior offenses)
- administrative information about the case (e.g., case number, arrest date, zipcode)

Improper modeling of domain knowledge and requirements:

- Use of irrelevant features
- Biased historical data



Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

VERNON PRATER

Prior Offenses
2 armed robberies, 1
attempted armed
robbery

Subsequent Offenses
1 grand theft

LOW RISK

3

BRISHA BORDEN

Prior Offenses
4 juvenile
misdemeanors

Subsequent Offenses
None

HIGH RISK

8

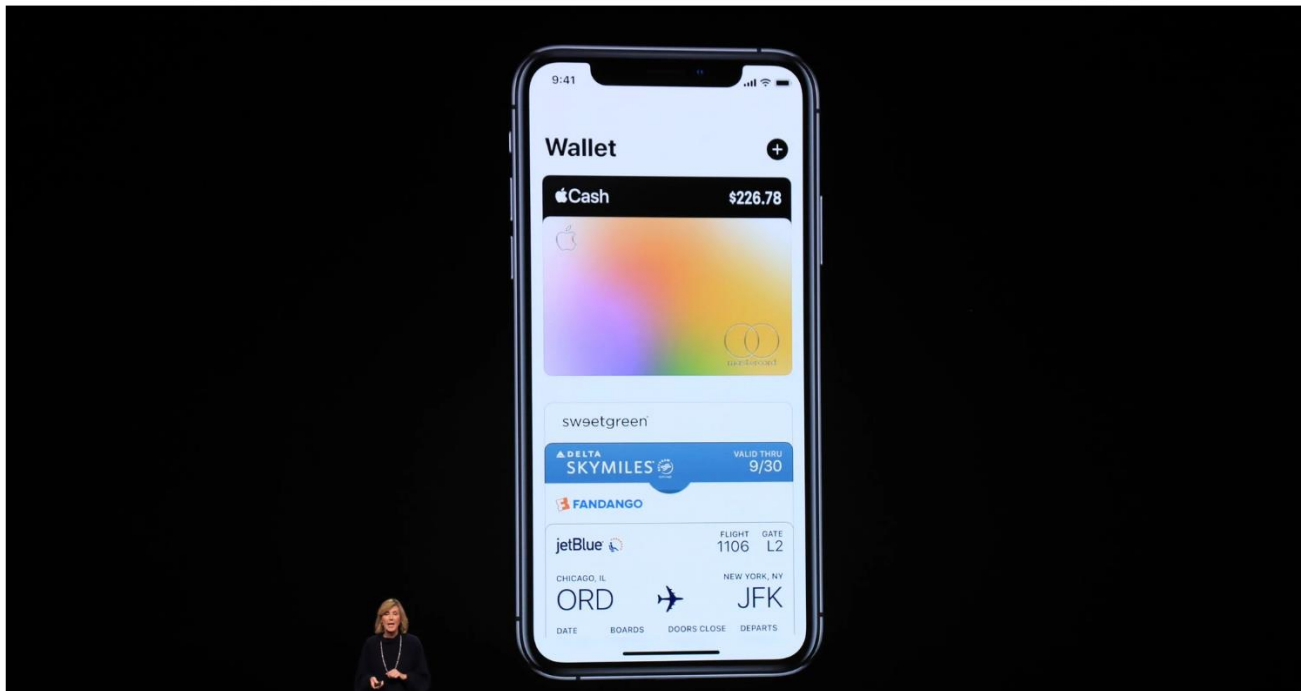
	White	African American
Did re-offend among the labeled higher risk	70%	71%
Labeled higher risk, but did not re-offend	23.5%	44.9%
Labeled lower risk, yet did re-offend	47.7%	28.0%

ine Bias

There's software used across the country to predict future criminals. And it's

Apple Card Investigated After Gender Discrimination Complaints

A prominent software developer said on Twitter that the credit card was “sexist” against women applying for credit.



- Lack of proper use-case analysis
 - Not up-to-date dataset
 - Incorrect feature values
 - e.g. loans from a married couple registered only under men
 - → Data bias

Amazon scraps secret AI recruiting tool that showed bias against women

Jeffrey Dastin

8 MIN READ



SAN FRANCISCO (Reuters) - Amazon.com Inc's ([AMZN.O](#)) machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.

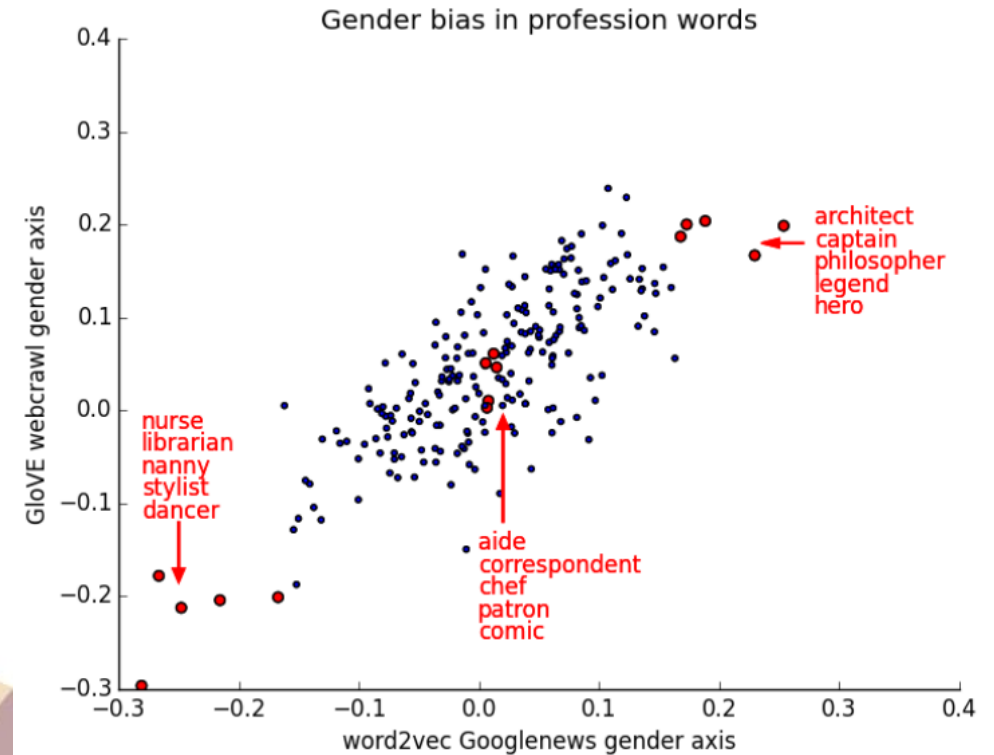
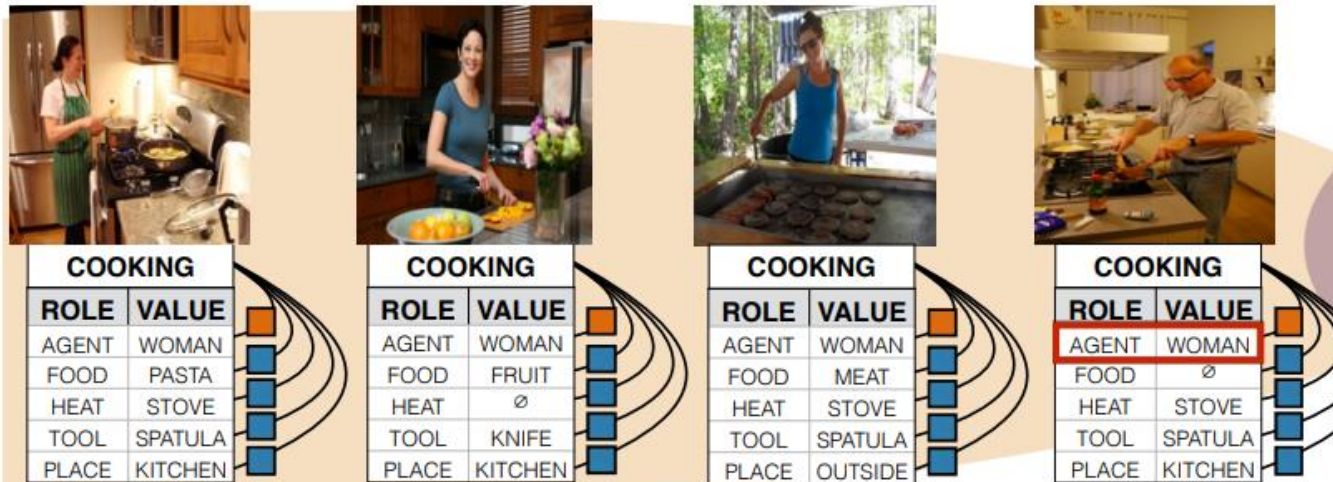
- Biased historical data:
 - Lack of data on the non-dominant group
 - Using an “objective” machine learning algorithm is not enough!
- Existence of proxy information for gender

Representational Bias

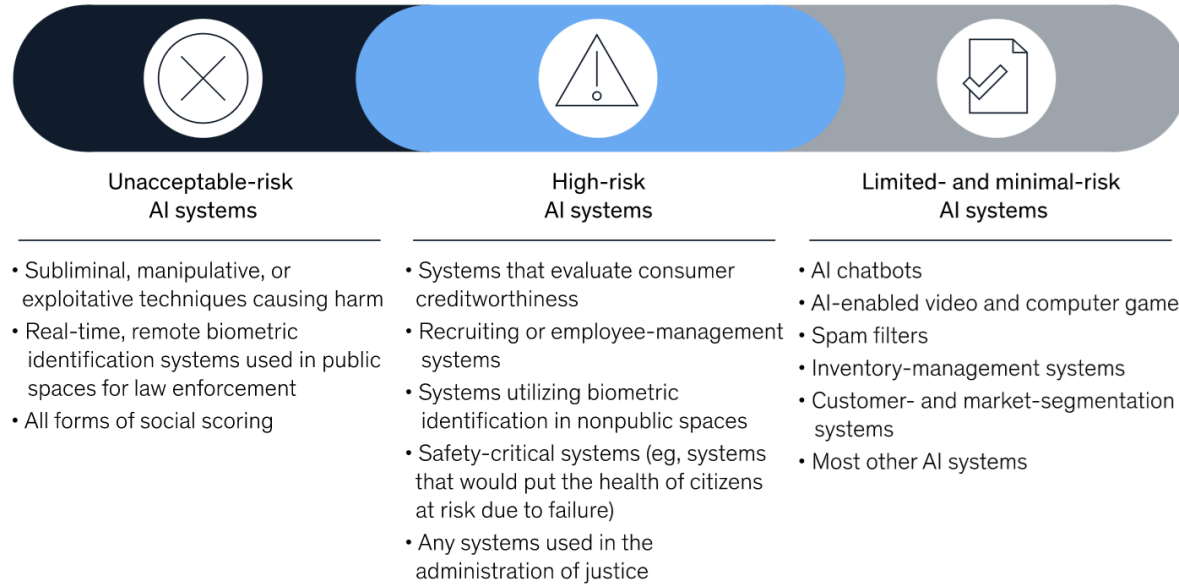
Embeddings are *highly* sexist!

Bolukbasi, T., Chang, K.W., Zou, J., Saligrama, V. and Kalai, A., 2016.

Quantifying and reducing stereotypes in word embeddings. *arXiv preprint*



The European Union's draft AI regulations classify AI systems into three risk categories.



The draft EU AI regulations place several requirements on organizations providing or using high-risk AI systems.

- | | |
|--|--|
| ☒ Implementation of a risk-management system | ☒ Human oversight |
| ☒ Data governance and management | ☒ Accuracy, robustness, and cybersecurity |
| ☒ Technical documentation | ☒ Conformity assessment |
| ☒ Record keeping and logging | ☒ Registration with EU-member-state government |
| ☒ Transparency and provision of information to users | ☒ Postmarket monitoring system |

<https://www.mckinsey.com/business-functions/mckinsey-analytics/our-insights/what-the-draft-european-union-ai-regulations-mean-for-business>

<https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1623335154975&uri=CELEX%3A52021PC0206>

ARTIFICIAL INTELLIGENCE



EUROPEAN COMMISSION

Brussels, 21.4.2021
COM(2021) 206 final
2021/0106(COD)

Proposal for a

REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL
LAYING DOWN HARMONISED RULES ON ARTIFICIAL INTELLIGENCE (ARTIFICIAL INTELLIGENCE ACT) AND AMENDING CERTAIN UNION LEGISLATIVE ACTS

{SEC(2021) 167 final} - {SWD(2021) 84 final} - {SWD(2021) 85 final}

EXPLANATORY MEMORANDUM

1. CONTEXT OF THE PROPOSAL

1.1. Reasons for and objectives of the proposal

This explanatory memorandum accompanies the proposal for a Regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Artificial Intelligence (AI) is a fast evolving family of technologies that can bring a wide array of economic and societal benefits across the entire spectrum of industries and social activities. By improving prediction, optimising operations and resource allocation, and personalising service delivery, the use of artificial intelligence can support socially and environmentally beneficial outcomes and provide key competitive advantages to companies and the European economy. Such action is especially needed in high-impact sectors, including climate change, environment and health, the public sector, finance, mobility, home affairs and agriculture. However, the same elements and techniques that power the socio-economic benefits of AI can also bring about new risks or negative consequences for individuals or the society. In light of the speed of technological change and possible challenges, the EU is committed to strive for a balanced approach. It is in the Union interest to preserve the EU's technological leadership and to ensure that Europeans can benefit from new technologies developed and functioning according to Union values, fundamental rights and principles.

This proposal delivers on the political commitment by President von der Leyen, who announced in her political guidelines for the 2019-2024 Commission "A Union that strives for more"¹, that the Commission would put forward legislation for a coordinated European approach on the human and ethical implications of AI. Following on that announcement, on 19 February 2020 the Commission published the White Paper on AI - A European approach to excellence and trust². The White Paper sets out policy options on how to achieve the twin objective of promoting the uptake of AI and of addressing the risks associated with certain uses of such technology. This proposal aims to implement the second objective for the development of an ecosystem of trust by promoting the development of trustworthy AI. The proposal is based on EU values and fundamental rights and principles.

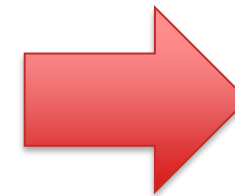
The Larger Issue?

- It's not just about bias, fairness and discrimination!

"Cow"



"Goat"



Magic
Machine
Learning
Model

“Cow”



“Goat”



training

Magic
Machine
Learning
Model

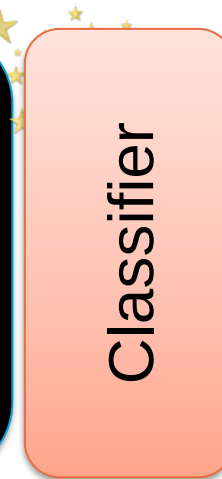
Classifier

classification



“Cow?”

“Cow”



classification



“Cow?”

“Cow”



“Goat”



training

Magic
Machine
Learning
Model

Classifier

classification

“Cow?”



“Cow”



“Goat”



training

**Magic
Machine
Learning
Model**

Classifier

classification

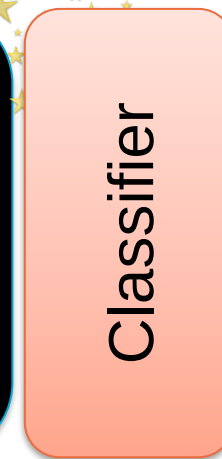


“Goat???”

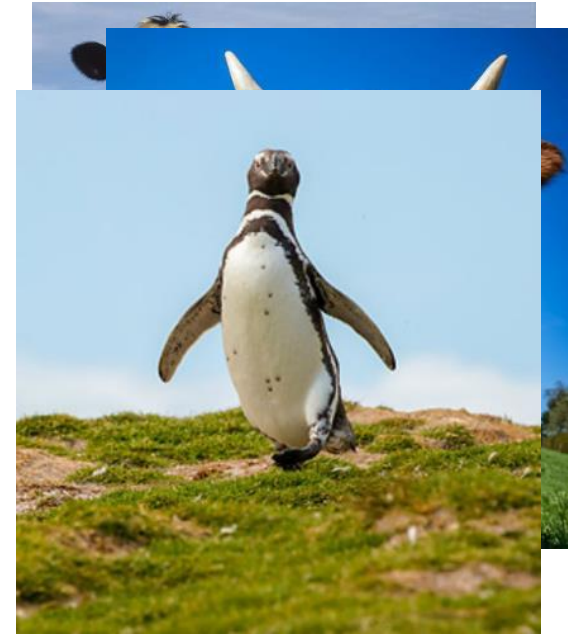
“Cow”



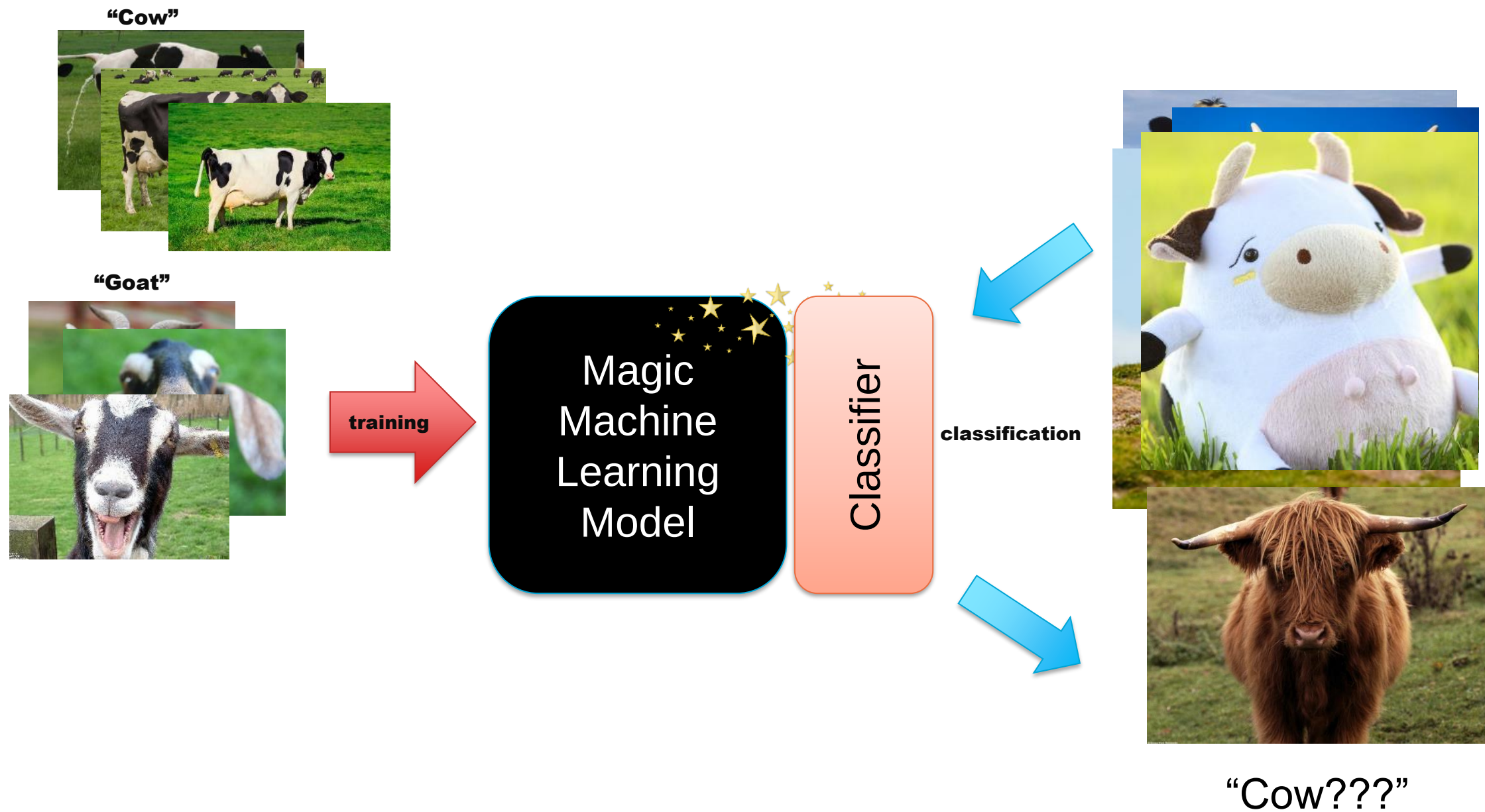
“Goat”

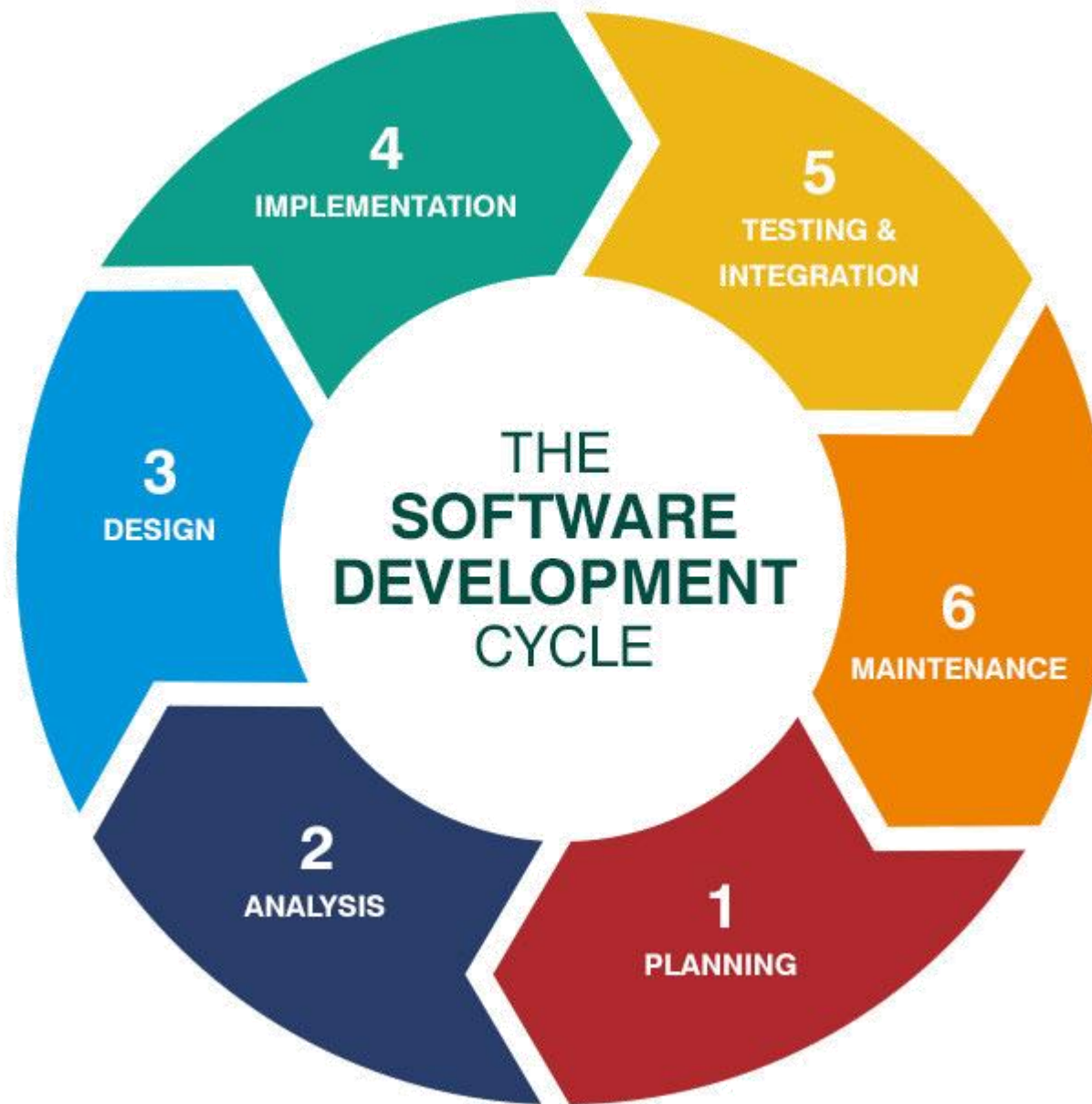


classification



“Goat???”





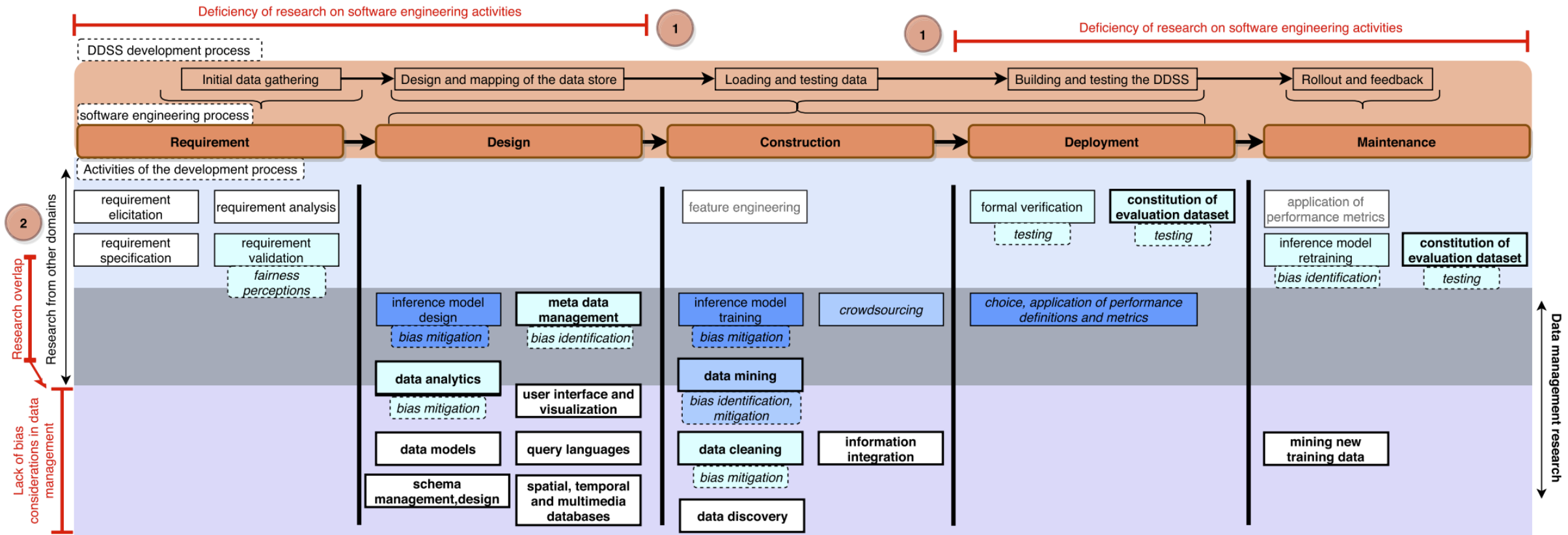


Fig. 2 Activities relevant to bias and their amount of bias-related research (white: no research; light to dark blue from few to plenty of research). Data management activities are bold

A.M.A. Balayn, C. Lofi, G.J.P.M. Houben (2021), [Managing bias and unfairness in data for decision support: a survey of machine learning and data engineering approaches to identify and mitigate bias and unfairness within data management and analytics systems](#), In *The VLDB Journal* Volume 30 p.739-768.

Bias-Aware Data Requirements



Bias Metrics



Bias-Aware Schema
Design

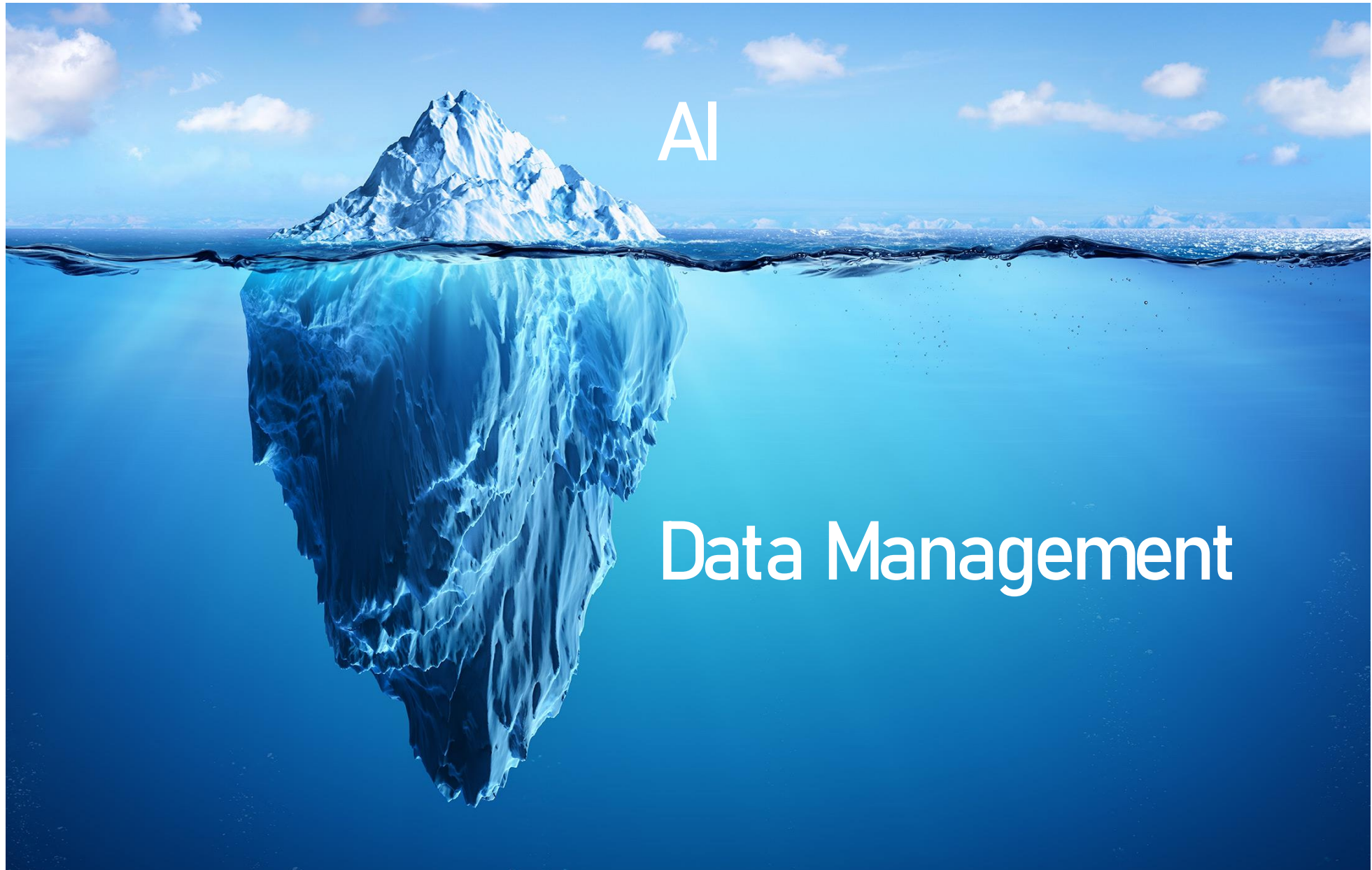


Bias Constraints



Bias Curation Methods

Integration in Data Management &
Engineering Systems?



Reliable AI

Machine
Learning

Data
Analytics

Data
Visualization

...

...

Solid Data Management

Domain Competence



THANKS!

Christoph Lofi
c.lofi@tudelft.nl